Geometry 1

Key definitions

Problem 1: Product of manifolds

Problem. If X and Y are manifolds, show that $X \times Y$ is a manifold and

$$\dim(X \times Y) = \dim X + \dim Y.$$

Theory / idea. If $\dim X = n$ and $\dim Y = m$, then the natural charts on $X \times Y$ are the *product charts*

$$(U \times V,\ \varphi \times \psi), \qquad (\varphi \times \psi)(x, y) = (\varphi(x), \psi(y)),$$

where (U, φ) is a chart on X and (V, ψ) is a chart on Y.

Solution. Let $\dim X = n$ and $\dim Y = m$.

1. **Hausdorff and second countable.** Since X and Y are Hausdorff and second countable, the product space $X \times Y$ (with the product topology) is also Hausdorff and second countable.

2. **Local Euclidean structure.** Fix charts (U, φ) on X and (V, ψ) on Y. Then $U \times V$ is open in $X \times Y$, and

$$\varphi \times \psi : U \times V \longrightarrow \varphi(U) \times \psi(V) \subset \mathbb{R}^n \times \mathbb{R}^m$$

is a homeomorphism. Moreover, $\varphi(U)$ and $\psi(V)$ are open, hence $\varphi(U) \times \psi(V)$ is open in $\mathbb{R}^{n+m}$ under the identification $\mathbb{R}^n \times \mathbb{R}^m \cong \mathbb{R}^{n+m}$. Thus every point of $X \times Y$ has a neighborhood homeomorphic to an open subset of $\mathbb{R}^{n+m}$.

3. **Smooth compatibility of charts.** Consider two product charts $(U_1 \times V_1, \varphi_1 \times \psi_1)$ and $(U_2 \times V_2, \varphi_2 \times \psi_2)$ with overlap. The transition map equals

$$(\varphi_2 \times \psi_2) \circ (\varphi_1 \times \psi_1)^{-1}(u, v) = \big(\varphi_2 \circ \varphi_1^{-1}(u),\ \psi_2 \circ \psi_1^{-1}(v)\big),$$

which is smooth because $\varphi_2 \circ \varphi_1^{-1}$ and $\psi_2 \circ \psi_1^{-1}$ are smooth.

Therefore $X \times Y$ is a smooth manifold, and the local model is $\mathbb{R}^{n+m}$, so $\dim(X \times Y) = n + m = \dim X + \dim Y$. $\square$

Why $\dim(\mathbb{R}^k \times \mathbb{R}^r) = k + r$ **(and not** kr**).** Be careful: there are two different "products" that look similar in notation.

(1) Cartesian product of Euclidean spaces. A point of $\mathbb{R}^k \times \mathbb{R}^r$ is a pair (x, y) where

$$x = (x_1, \dots, x_k) \in \mathbb{R}^k, \qquad y = (y_1, \dots, y_r) \in \mathbb{R}^r.$$

Thus (x, y) is described by $k + r$ real coordinates, and there is a natural linear diffeomorphism

$$\mathbb{R}^k \times \mathbb{R}^r \;\cong\; \mathbb{R}^{k+r}, \qquad (x_1, \dots, x_k,\; y_1, \dots, y_r) \longleftrightarrow (x, y).$$

Hence

$$\dim(\mathbb{R}^k \times \mathbb{R}^r) = k + r.$$

(2) Where the product kr **comes from.** The number kr appears for the *space of* $k \times r$ *matrices*:

$$\mathrm{Mat}_{k\times r}(\mathbb{R}) \;\cong\; \mathbb{R}^{kr},$$

because a $k \times r$ matrix has kr independent entries. Likewise, for finite-dimensional vector spaces V, W one has

$$\dim(V \otimes W) = \dim(V) \cdot \dim(W),$$

which is a different construction than the Cartesian product.

Connection to manifolds. If $\dim X = n$ and $\dim Y = m$, then locally $X \approx \mathbb{R}^n$ and $Y \approx \mathbb{R}^m$, so locally

$$X \times Y \approx \mathbb{R}^n \times \mathbb{R}^m \cong \mathbb{R}^{n+m},$$

which explains why $\dim(X \times Y) = n + m$.

Products of manifolds

Cartesian products and dimension. For $k, r \in \mathbb{N}$, the Cartesian product $\mathbb{R}^k \times \mathbb{R}^r$ consists of pairs (x, y) with $x \in \mathbb{R}^k$ and $y \in \mathbb{R}^r$. Define

$$\Phi : \mathbb{R}^k \times \mathbb{R}^r \to \mathbb{R}^{k+r}, \qquad \Phi(x, y) = (x_1, \dots, x_k, y_1, \dots, y_r).$$

Then Φ is a linear isomorphism, hence a homeomorphism and diffeomorphism. Therefore

$$\dim(\mathbb{R}^k \times \mathbb{R}^r) = k + r.$$

The number kr appears instead as $\dim(\mathbb{R}^k \otimes \mathbb{R}^r) = kr$ (tensor product).

Product manifolds. Let M and N be smooth manifolds of dimensions m and n. Then $M \times N$ carries a natural smooth manifold structure of dimension $m + n$.

Product charts. If (U, φ) is a chart on M with $\varphi : U \to \mathbb{R}^m$ and (V, ψ) is a chart on N with $\psi : V \to \mathbb{R}^n$, define the product chart

$$(U \times V,\; \varphi \times \psi), \qquad (\varphi \times \psi)(p, q) = (\varphi(p), \psi(q)) \in \mathbb{R}^{m+n}.$$

The collection of such product charts forms an atlas on $M \times N$ and the transition maps are smooth, so $M \times N$ is a smooth manifold with $\dim(M \times N) = m + n$.

Canonical maps. The projections

$$\pi_1 : M \times N \to M, \quad \pi_1(p,q) = p, \qquad \pi_2 : M \times N \to N, \quad \pi_2(p,q) = q$$

are smooth. In product coordinates $(x,y) \in \mathbb{R}^{m+n}$ they are given by $\pi_1(x,y) = x$ and $\pi_2(x,y) = y$, hence their differentials are surjective everywhere, so π_1 and π_2 are submersions.

Fix $q_0 \in N$. The map

$$i_{q_0} : M \to M \times N, \quad i_{q_0}(p) = (p, q_0)$$

is smooth and in coordinates is $x \mapsto (x, \mathrm{const})$, hence di_{q_0} has rank m everywhere, so i_{q_0} is an immersion.

The flip map

$$\tau : M \times N \to N \times M, \quad \tau(p,q) = (q,p)$$

is a diffeomorphism with inverse itself.

Problem 2: A diffeomorphism from a ball onto $\mathbb{R}^k$

Problem. Let $B_r = \{x \in \mathbb{R}^k : \|x\| < r\}$ be the open ball. Show that the map

$$F : B_r \longrightarrow \mathbb{R}^k, \qquad F(x) = \frac{r\,x}{\sqrt{r^2 - \|x\|^2}}$$

is a diffeomorphism of B_r onto $\mathbb{R}^k$.

Solution. **(1) Smoothness of F.** On B_r we have $r^2 - \|x\|^2 > 0$, hence $\sqrt{r^2 - \|x\|^2}$ is a smooth function of x. Thus F is smooth on B_r.

(2) Construct the inverse map. Given $y = F(x)$, we compute

$$\|y\|^2 = \frac{r^2\|x\|^2}{r^2 - \|x\|^2}.$$

Solving for $\|x\|^2$ gives

$$\|x\|^2 = \frac{r^2\|y\|^2}{r^2 + \|y\|^2}.$$

Then

$$r^2 - \|x\|^2 = r^2 - \frac{r^2\|y\|^2}{r^2 + \|y\|^2} = \frac{r^4}{r^2 + \|y\|^2}, \qquad \sqrt{r^2 - \|x\|^2} = \frac{r^2}{\sqrt{r^2 + \|y\|^2}}.$$

From $y = \dfrac{rx}{\sqrt{r^2 - \|x\|^2}}$ we obtain

$$y = \frac{rx}{r^2/\sqrt{r^2 + \|y\|^2}} = x \cdot \frac{\sqrt{r^2 + \|y\|^2}}{r} \quad \Longrightarrow \quad x = \frac{r\,y}{\sqrt{r^2 + \|y\|^2}}.$$

Define

$$G : \mathbb{R}^k \to \mathbb{R}^k, \qquad G(y) = \frac{r\,y}{\sqrt{r^2 + \|y\|^2}}.$$

We have $\|G(y)\|^2 = \dfrac{r^2\|y\|^2}{r^2+\|y\|^2} < r^2$, so $G(y) \in B_r$ for all y. Moreover, a direct substitution shows

$$F(G(y)) = y \quad (\forall y \in \mathbb{R}^k), \qquad G(F(x)) = x \quad (\forall x \in B_r).$$

Thus $G = F^{-1}$.

(3) Smoothness of the inverse. The map G is smooth on all of $\mathbb{R}^k$ because $r^2 + \|y\|^2 > 0$ everywhere. Hence F is bijective, smooth, and has a smooth inverse: F is a diffeomorphism $B_r \cong \mathbb{R}^k$. □

Remark. If a chart maps a neighborhood in a manifold onto a ball $B_r \subset \mathbb{R}^k$, then composing with the above diffeomorphism $B_r \to \mathbb{R}^k$ yields a chart whose coordinate domain is all of $\mathbb{R}^k$.

How this implies charts can be chosen with domain $\mathbb{R}^k$. Let M be a smooth k-manifold and let (U, φ) be a chart, so $\varphi : U \to \varphi(U) \subset \mathbb{R}^k$ is a diffeomorphism onto an open set. Fix $p \in U$ and set $a = \varphi(p)$.

Since $\varphi(U)$ is open, there exists $r > 0$ such that the Euclidean ball $B_r(a) \subset \varphi(U)$. Let $T_{-a}(x) = x - a$ be translation by $-a$ and define

$$\psi := T_{-a} \circ \varphi : U \longrightarrow \varphi(U) - a.$$

Then ψ is a diffeomorphism, $\psi(p) = 0$, and $B_r(0) \subset \psi(U)$.

Now set

$$U' := \psi^{-1}(B_r(0)) \subset U.$$

Then $\psi|_{U'} : U' \to B_r(0)$ is a diffeomorphism.

By Problem 2, the map

$$F : B_r(0) \longrightarrow \mathbb{R}^k, \qquad F(x) = \frac{r\,x}{\sqrt{r^2 - \|x\|^2}}$$

is a diffeomorphism. Therefore the composition

$$\Phi := F \circ \psi|_{U'} : U' \longrightarrow \mathbb{R}^k$$

is a diffeomorphism onto all of $\mathbb{R}^k$. Hence, after possibly shrinking the neighborhood, one may always choose local coordinates whose coordinate domain is the whole space $\mathbb{R}^k$.

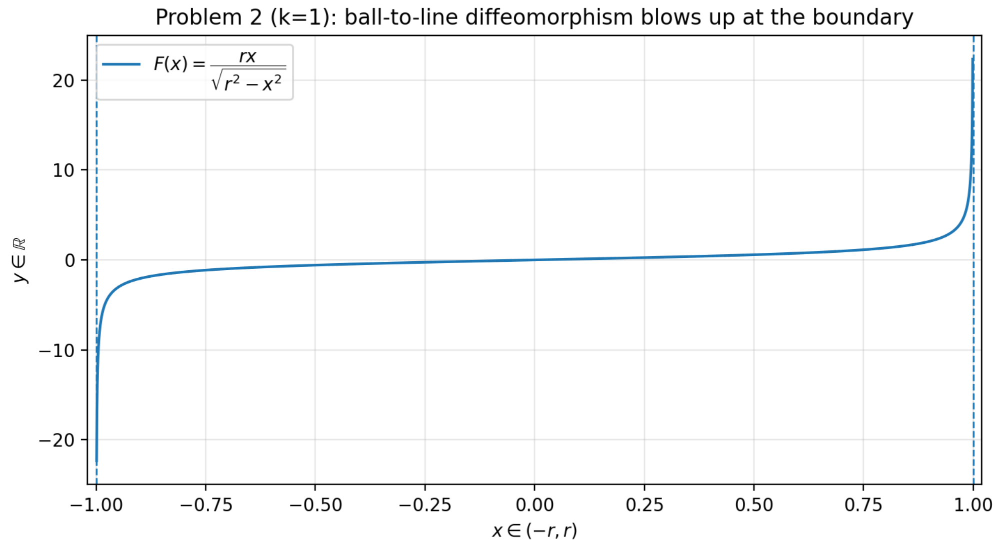


Figure 1: (Problem 2, $k = 1$, $r = 1$) The map $F(x) = \dfrac{rx}{\sqrt{r^2 - x^2}}$ blows up as $x \to \pm r$, so the boundary of the interval is sent to infinity.

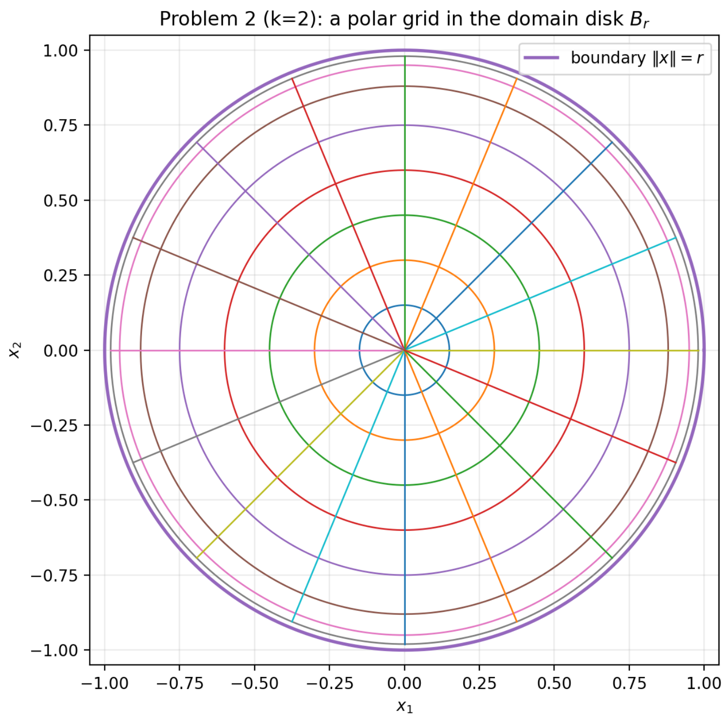


Figure 2: (Problem 2, $k = 2$, $r = 1$) A polar grid in the domain disk B_r. Circles $\|x\| = \rho$ and rays $\theta = \text{const}$ are shown.

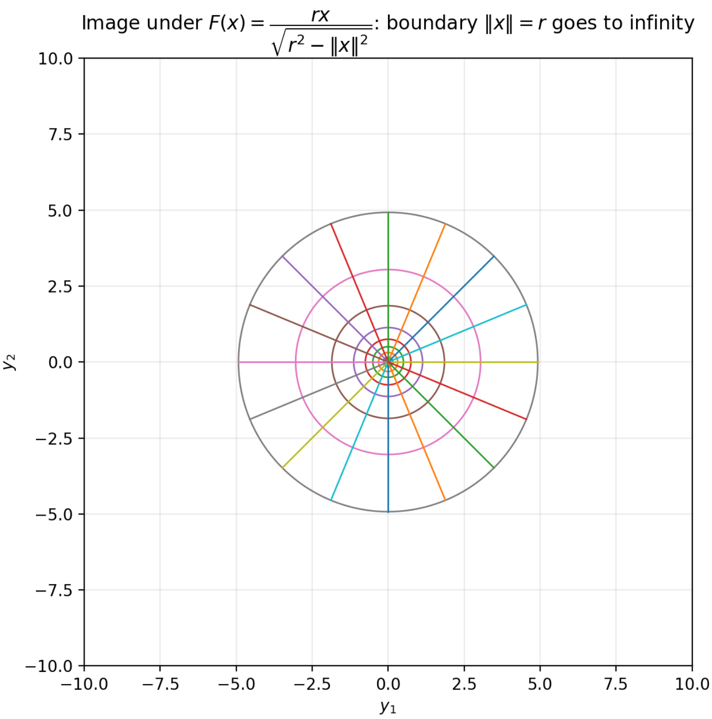


Figure 3: Image of the grid under $F(x) = \dfrac{rx}{\sqrt{r^2 - \|x\|^2}}$. Rays stay rays and circles map to circles, while the boundary $\|x\| = r$ is pushed out to infinity.

Problem 3: Open subsets of $\mathbb{R}^n$ and $\mathbb{R}^m$ with $n \neq m$ are not diffeomorphic

Problem. If $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ are open subsets with $n \neq m$, prove that U and V are not diffeomorphic.

Solution. Assume for contradiction that there exists a diffeomorphism $f : U \to V$. Then $f^{-1} : V \to U$ is smooth and

$$f^{-1} \circ f = \mathrm{id}_U, \qquad f \circ f^{-1} = \mathrm{id}_V.$$

Fix $p \in U$ and set $q = f(p) \in V$. Differentiating and using the chain rule,

$$D(f^{-1})_q \circ Df_p = I_{\mathbb{R}^n}, \qquad Df_p \circ D(f^{-1})_q = I_{\mathbb{R}^m}.$$

Thus $Df_p : \mathbb{R}^n \to \mathbb{R}^m$ has both a left-inverse and a right-inverse, so it is a linear isomorphism. But a linear isomorphism $\mathbb{R}^n \cong \mathbb{R}^m$ exists only if $n = m$. This contradicts $n \neq m$. Therefore no diffeomorphism exists. $\square$

Problem 4

(i) Is the union of the two coordinate axes in $\mathbb{R}^2$ a manifold? Let

$$X := \{(x, 0) : x \in \mathbb{R}\} \ \cup \ \{(0, y) : y \in \mathbb{R}\} \subset \mathbb{R}^2.$$

Away from the origin, X is locally a straight line, hence looks like a 1-manifold. The obstruction is at $(0, 0)$.

In a 1-manifold (with or without boundary), there exists a neighborhood U of any point p such that $U \setminus \{p\}$ has either 2 connected components (if p is an interior point, modeled on $(-1, 1)$) or 1 connected component (if p is a boundary point, modeled on $[0, 1)$).

However, for any neighborhood U of $(0, 0)$ in the subspace topology on X, the punctured set $U \setminus \{(0, 0)\}$ has *four* connected components (corresponding to the four arms $x > 0$, $x < 0$, $y > 0$, $y < 0$). Therefore X cannot be a 1-manifold (with or without boundary). Hence the union of the two coordinate axes is **not** a manifold. $\square$

(ii) The hyperboloid $x^2 + y^2 - z^2 = a$ is a manifold for $a > 0$; what about $a = 0$? Define

$$F : \mathbb{R}^3 \to \mathbb{R}, \qquad F(x, y, z) = x^2 + y^2 - z^2,$$

so the hyperboloid is $H_a := F^{-1}(a)$. We have

$$\nabla F(x, y, z) = (2x, 2y, -2z),$$

which vanishes only at $(0, 0, 0)$. If $a > 0$, then $(0, 0, 0) \notin H_a$, hence $\nabla F \neq 0$ everywhere on H_a. Thus a is a regular value of F, so H_a is a smooth submanifold of $\mathbb{R}^3$ of codimension 1, i.e. $\dim H_a = 2$.

For $a = 0$ we obtain the double cone $x^2 + y^2 = z^2$. Away from the origin the gradient is still nonzero, so it is a smooth 2-manifold there, but at $(0, 0, 0)$ the gradient vanishes, hence the origin is a singular point and the cone is **not** a manifold globally.

Tangent space at $(\sqrt{a}, 0, 0)$ **(with** $a > 0$**).** For a regular level set $F^{-1}(a)$ and $p \in F^{-1}(a)$,

$$T_pF^{-1}(a) = \{v \in \mathbb{R}^3 : \nabla F(p) \cdot v = 0\}.$$

At $p = (\sqrt{a}, 0, 0)$, $\nabla F(p) = (2\sqrt{a}, 0, 0)$, hence the condition is $2\sqrt{a}\, v_1 = 0$, so $v_1 = 0$. Therefore

$$T_{(\sqrt{a},0,0)}H_a = \{(0, v_2, v_3) : v_2, v_3 \in \mathbb{R}\},$$

the plane parallel to the yz-plane. $\square$

(iii) The solid hyperboloid $x^2 + y^2 - z^2 \leq a$ **is a manifold with boundary** $(a > 0)$**.** Let

$$W_a := \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 - z^2 \leq a\} = F^{-1}((-\infty, a]).$$

Then its boundary is

$$\partial W_a = F^{-1}(a) = H_a.$$

For $a > 0$ we have $\nabla F \neq 0$ on H_a, hence H_a is a smooth hypersurface. A standard regular-sublevel-set argument implies that W_a is a smooth 3-manifold with boundary, with interior $\{F < a\}$ and boundary $\{F = a\}$. $\square$

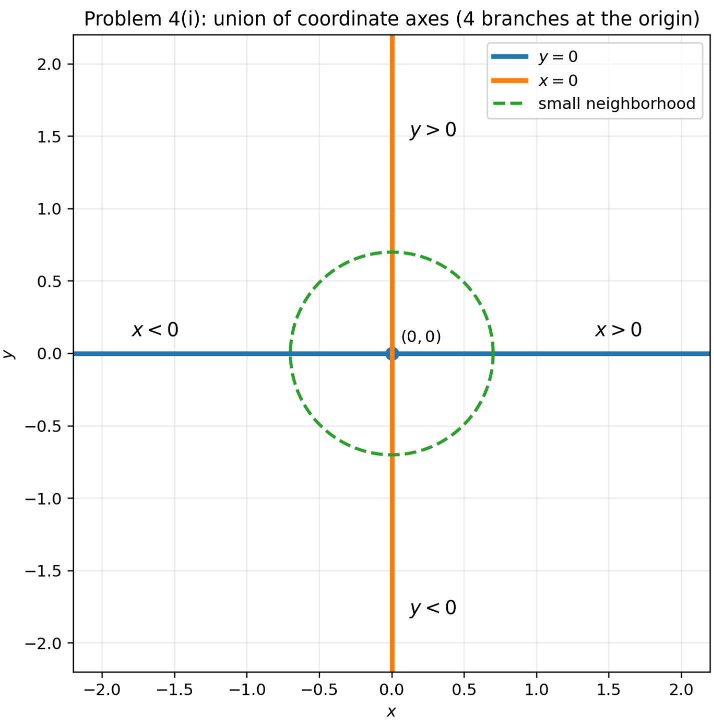


Figure 4: Problem 4(i): the union of coordinate axes. Any punctured neighborhood of $(0, 0)$ has four connected components (four "arms"), so it cannot be a 1-manifold (with or without boundary).

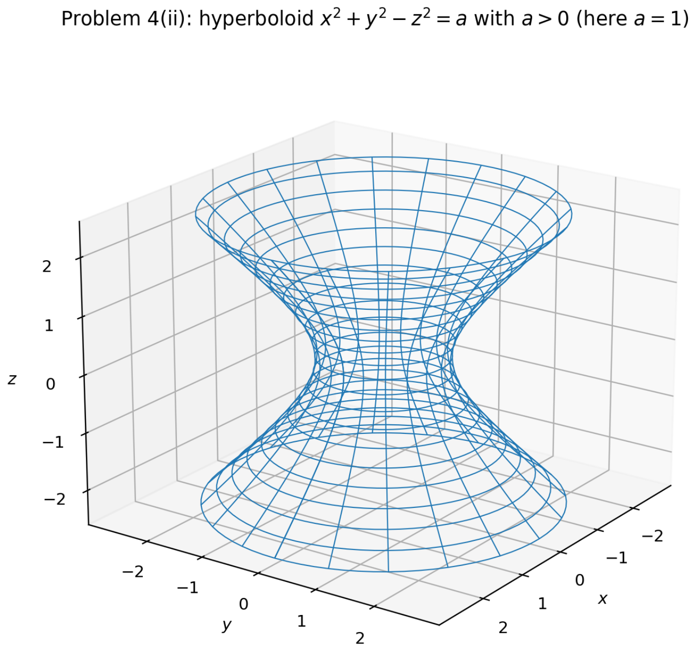


Figure 5: Problem 4(ii): the hyperboloid $x^2 + y^2 - z^2 = a$ for $a > 0$ (here $a = 1$). This is a smooth 2-dimensional submanifold of $\mathbb{R}^3$.

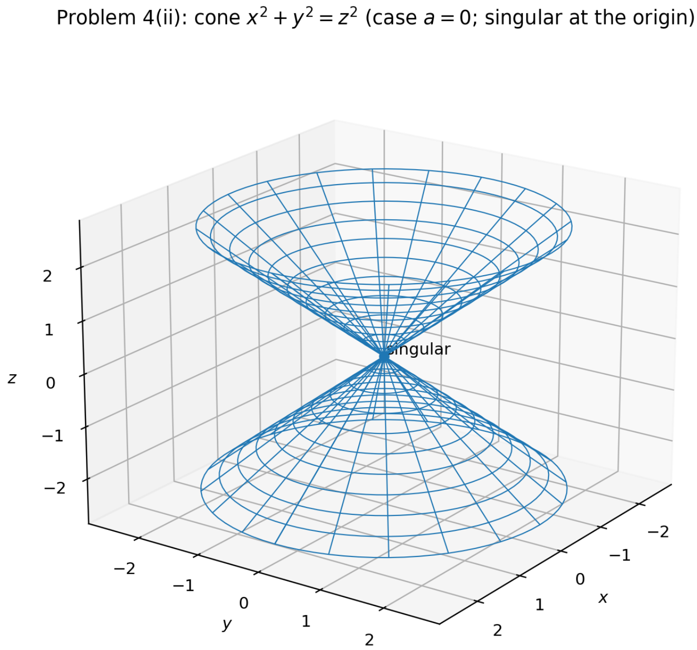


Figure 6: Problem 4(ii): the cone $x^2 + y^2 = z^2$ (case $a = 0$). It is smooth away from the origin, but the origin is a singular point, so it is not a manifold globally.

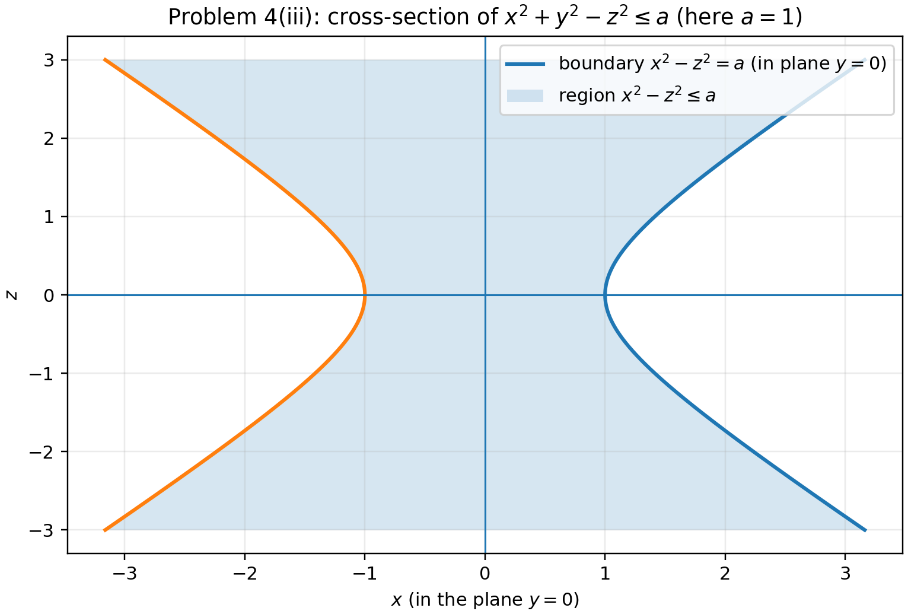


Figure 7: Problem 4(iii): cross-section ($y = 0$) of the sublevel set $x^2 + y^2 - z^2 \leq a$ (here $a = 1$). The boundary corresponds to $x^2 + y^2 - z^2 = a$.

Remark (hyperboloid vs. cone as $a \to 0$). Both surfaces come from the same level-set family of the function

$$F(x, y, z) = x^2 + y^2 - z^2, \qquad H_a := F^{-1}(a) = \{(x, y, z) \in \mathbb{R}^3 : \; x^2 + y^2 - z^2 = a\}.$$

For $a > 0$ the set H_a is a smooth 2-dimensional submanifold (a hyperboloid-type surface). If $a = 0$ we obtain

$$H_0 = \{x^2 + y^2 - z^2 = 0\} = \{x^2 + y^2 = z^2\},$$

which is exactly the *double cone*. Thus the cone is the $a = 0$ member of the same family, and one may view H_a as degenerating to the cone as $a \downarrow 0$.

A geometric way to see this is via horizontal slices: fixing z,

$$x^2 + y^2 = a + z^2.$$

Hence, for $a > 0$ the slice $H_a \cap \{z = \text{const}\}$ is a circle of radius $\sqrt{a + z^2}$; in particular at $z = 0$ the radius is $\sqrt{a}$. As $a \to 0^+$ this "waist" radius tends to 0, and the surface collapses to the cone $x^2 + y^2 = z^2$.

The manifold property breaks precisely at $a = 0$ because

$$\nabla F(x, y, z) = (2x, 2y, -2z).$$

For $a > 0$, $(0, 0, 0) \notin H_a$, so $\nabla F \neq 0$ everywhere on H_a and the level set is regular. For $a = 0$, the origin lies on H_0 and $\nabla F(0, 0, 0) = 0$, so the origin is a singular point; therefore H_0 is not a manifold globally (though it is smooth away from the origin).

Problem 5

Problem. Show that the set of all 2×2 matrices of rank 1 is a 3-dimensional submanifold of $\mathbb{R}^4$.

Solution. Identify $\mathbb{R}^4$ with matrices

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Rank 1 means $\det(A) = 0$ but $A \neq 0$. Define the smooth function

$$G : \mathbb{R}^4 \to \mathbb{R}, \qquad G(a, b, c, d) = ad - bc.$$

Then

$$\{A : \operatorname{rank}(A) = 1\} = \{G = 0\} \setminus \{0\}.$$

Compute

$$\nabla G(a, b, c, d) = (d,\ -c,\ -b,\ a).$$

If $\nabla G = 0$, then $a = b = c = d = 0$, i.e. only at the zero matrix. Therefore, on $\{G = 0\} \setminus \{0\}$, the gradient is never zero, so 0 is a regular value (of G restricted to $\mathbb{R}^4 \setminus \{0\}$). Hence $\{G = 0\} \setminus \{0\}$ is a smooth embedded submanifold of codimension 1 in $\mathbb{R}^4$, so it has dimension $4 - 1 = 3$. $\square$

Problem 6 (Local immersion theorem)

Problem. The canonical immersion is the standard inclusion $\mathbb{R}^k \hookrightarrow \mathbb{R}^n$ for $k \leq n$,

$$(x_1, \ldots, x_k) \longmapsto (x_1, \ldots, x_k, 0, \ldots, 0).$$

Let f be an immersion, $y = f(x)$. Show that there exist local coordinates around x and y such that f in these coordinates is the canonical immersion.

Solution. Choose charts (U, φ) about $x \in M$ and (V, ψ) about $y = f(x) \in N$ such that $\varphi(x) = 0 \in \mathbb{R}^k$ and $\psi(y) = 0 \in \mathbb{R}^n$. In coordinates we obtain a smooth map

$$\tilde{f} := \psi \circ f \circ \varphi^{-1} : \varphi(U) \subset \mathbb{R}^k \longrightarrow \mathbb{R}^n$$

with $\tilde{f}(0) = 0$. Since f is an immersion at x, the differential $D\tilde{f}(0)$ has rank k.

Let $\pi : \mathbb{R}^n \to \mathbb{R}^k$ be the projection onto the first k coordinates and set

$$g := \pi \circ \tilde{f} : \varphi(U) \to \mathbb{R}^k.$$

Because $\operatorname{rank}(D\tilde{f}(0)) = k$, after possibly reordering target coordinates we may assume $Dg(0)$ is invertible. By the inverse function theorem, g is a local diffeomorphism near 0. Thus we may use $u = g(x)$ as new local coordinates on the domain (shrink U if needed).

In these coordinates, $\tilde{f}$ has the form

$$\tilde{f}(u) = (u, h(u))$$

for some smooth map h into $\mathbb{R}^{n-k}$ (split $\mathbb{R}^n \cong \mathbb{R}^k \times \mathbb{R}^{n-k}$). Now define a smooth change of coordinates on the target by

$$\Phi : \mathbb{R}^k \times \mathbb{R}^{n-k} \to \mathbb{R}^k \times \mathbb{R}^{n-k}, \qquad \Phi(u, w) = (u, w - h(u)).$$

This is a local diffeomorphism near $(0, 0)$, with inverse $(u, w) \mapsto (u, w + h(u))$. Then

$$(\Phi \circ \tilde{f})(u) = \Phi(u, h(u)) = (u, 0),$$

which is exactly the canonical immersion $\mathbb{R}^k \hookrightarrow \mathbb{R}^n$ in these coordinates. Translating back to manifold language gives the desired local coordinate systems around x and y. $\square$

Immersion vs. embedding (important distinction). Let $f : M \to N$ be a smooth map between smooth manifolds, $\dim M = k$, $\dim N = n$.

- f is an **immersion** if for every $x \in M$ the differential

$$df_x : T_xM \to T_{f(x)}N$$

is injective (equivalently $\operatorname{rank}(df_x) = k$). In particular, $k \leq n$.

- f is a **(smooth) embedding** if it is an immersion and a homeomorphism onto its image $f(M) \subset N$ (with the subspace topology). In particular, an embedding is injective.

Thus:

$$\text{embedding} \implies \text{immersion}, \qquad \text{but not conversely in general.}$$

Local picture (link to the local immersion theorem). The local immersion theorem says that if f is an immersion at x, then there exist local coordinates around x and $y = f(x)$ in which f becomes the canonical inclusion

$$(u_1, \ldots, u_k) \longmapsto (u_1, \ldots, u_k, 0, \ldots, 0).$$

In particular, *every immersion is locally an embedding*, but it can fail to be an embedding globally (e.g. by self-intersections or lack of injectivity).

Characteristic examples.

1. **Embedding.** The standard inclusion $S^1 \hookrightarrow \mathbb{R}^2$ given by

$$f(e^{it}) = (\cos t, \sin t)$$

is an embedding (injective and a homeomorphism onto its image).

2. **Immersion but not embedding (self-intersection).** The map $f : S^1 \to \mathbb{R}^2$ defined by

$$f(t) = (\sin t, \sin 2t)$$

has $f'(t) = (\cos t, 2\cos 2t) \neq (0, 0)$ for all t, hence it is an immersion. However, it has a self-intersection (a double point), so it is not injective and therefore not an embedding.

3. **Immersion but not embedding (non-injective covering).** The map $f : \mathbb{R} \to \mathbb{R}^2$ given by

$$f(t) = (\cos t, \sin t)$$

is an immersion since $f'(t) = (-\sin t, \cos t) \neq (0, 0)$ for all t, but it is not injective (e.g. $f(0) = f(2\pi)$), hence not an embedding.

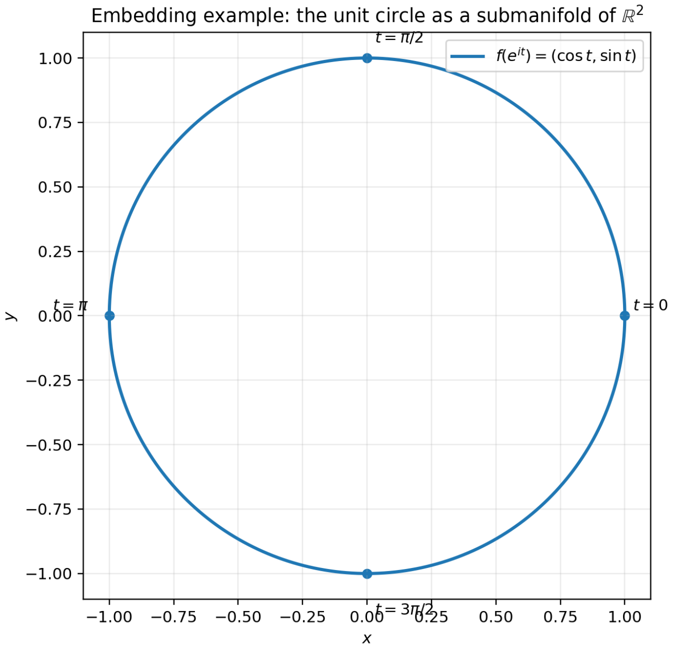


Figure 8: Embedding example: $S^1 \hookrightarrow \mathbb{R}^2$, $f(e^{it}) = (\cos t, \sin t)$.

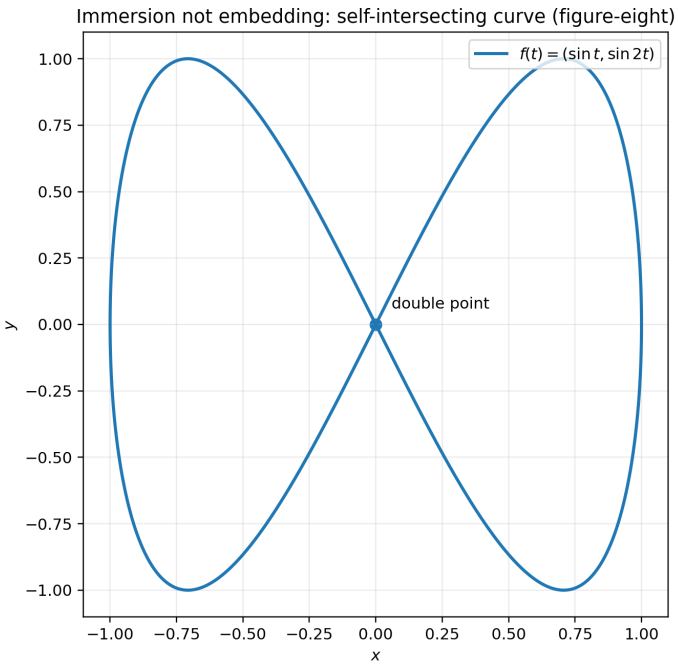


Figure 9: Immersion not embedding: $f(t) = (\sin t, \sin 2t)$ has a self-intersection (double point).

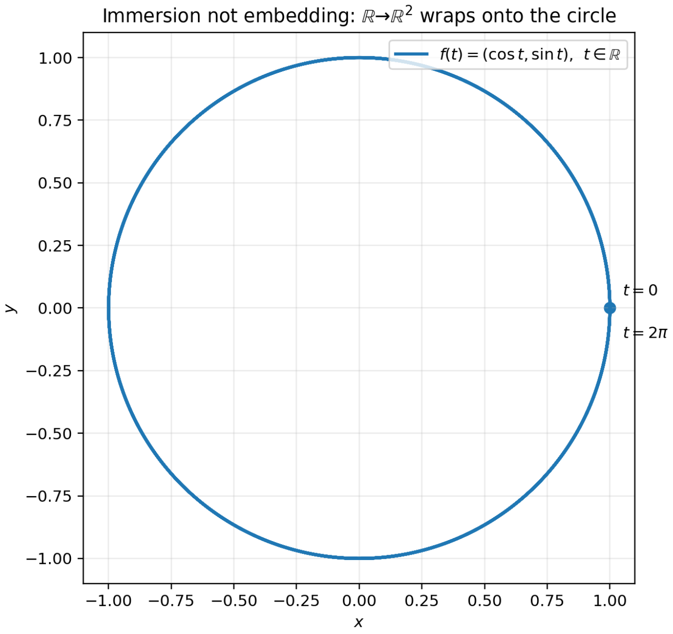


Figure 10: Immersion not embedding: $f : \mathbb{R} \to \mathbb{R}^2$, $f(t) = (\cos t, \sin t)$ is not injective.

Problem 7 (Local submersion theorem and consequences)

Problem statement. The *canonical submersion* is the standard projection $\mathbb{R}^k \to \mathbb{R}^\ell$ for $k \geq \ell$,

$$(x_1, \dots, x_k) \longmapsto (x_1, \dots, x_\ell).$$

Let f be a submersion and $y = f(x)$. Show:

1. there exist local coordinates around x and y such that f becomes the canonical submersion;
2. submersions are open maps;
3. if X is compact and Y connected, then every submersion $f : X \to Y$ is surjective;
4. decide whether there exist submersions from compact manifolds into Euclidean spaces.

Theory / key fact. A smooth map $f : X^k \to Y^\ell$ is a *submersion at x* if $df_x : T_xX \to T_{f(x)}Y$ is surjective. Equivalently, $\operatorname{rank}(df_x) = \ell$ (so $\ell \leq k$). The constant rank theorem gives a local normal form.

(i) Local normal form. Since f is a submersion at x, we have $\operatorname{rank}(df_x) = \ell$. By the constant rank theorem, there exist coordinate charts (U, φ) about x and (V, ψ) about $y = f(x)$ such that in these coordinates

$$(\psi \circ f \circ \varphi^{-1})(u_1, \dots, u_k) = (u_1, \dots, u_\ell),$$

i.e. f becomes the canonical projection $\mathbb{R}^k \to \mathbb{R}^\ell$. $\square$

(ii) Submersions are open maps. Let $U \subset X$ be open. Fix $x \in U$ and choose local coordinates as in (i), so that locally f is the projection $\pi(u_1, \ldots, u_k) = (u_1, \ldots, u_\ell)$. The projection $\pi : \mathbb{R}^k \to \mathbb{R}^\ell$ is an open map. Since coordinate changes are diffeomorphisms (hence homeomorphisms), they preserve openness. Therefore $f(U)$ is open in Y. Thus every submersion is an open map. $\square$

(iii) Compact X and connected Y imply surjectivity. Let $f : X \to Y$ be a submersion. Since X is compact and f is continuous, $f(X)$ is compact. Because manifolds are Hausdorff, compact sets are closed, hence $f(X)$ is closed in Y. On the other hand, by (ii), $f(X) = f(\,X\,)$ is open in Y (since X is open in itself). Assuming $X \neq \emptyset$, the set $f(X)$ is nonempty. Hence $f(X)$ is a nonempty subset of Y that is both open and closed. If Y is connected, this forces $f(X) = Y$. Thus f is surjective. $\square$

(iv) Submersions into Euclidean spaces from compact manifolds. Let X be compact and suppose $f : X \to \mathbb{R}^\ell$ is a submersion with $\ell \geq 1$. Then $\mathbb{R}^\ell$ is connected, so by (iii) the map must be surjective: $f(X) = \mathbb{R}^\ell$. But $f(X)$ is compact, hence cannot equal the non-compact space $\mathbb{R}^\ell$. This contradiction shows that no such submersion exists for $\ell \geq 1$. (For $\ell = 0$, the target is a point and any smooth map is trivially a submersion.) $\square$

Normal forms (constant rank theorem viewpoint). A *normal form* is the expression of a smooth map after choosing suitable local coordinates in the domain and codomain.

- **Immersion normal form.** If $f : M^k \to N^n$ is an immersion at x (so df_x is injective, rank k), then there exist local coordinates u near x and v near $y = f(x)$ such that

  $$(v \circ f \circ u^{-1})(u_1, \ldots, u_k) = (u_1, \ldots, u_k, 0, \ldots, 0).$$

  Geometrically, $f(M)$ looks locally like a coordinate k-plane in N.

- **Submersion normal form.** If $f : X^k \to Y^\ell$ is a submersion at x (so df_x is surjective, rank ℓ), then there exist local coordinates u near x and v near $y = f(x)$ such that

  $$(v \circ f \circ u^{-1})(u_1, \ldots, u_k) = (u_1, \ldots, u_\ell).$$

  Geometrically, the fibers $f^{-1}(c)$ are locally the coordinate slices $\{u_1 = c_1, \ldots, u_\ell = c_\ell\}$, hence smooth manifolds of dimension $k - \ell$.

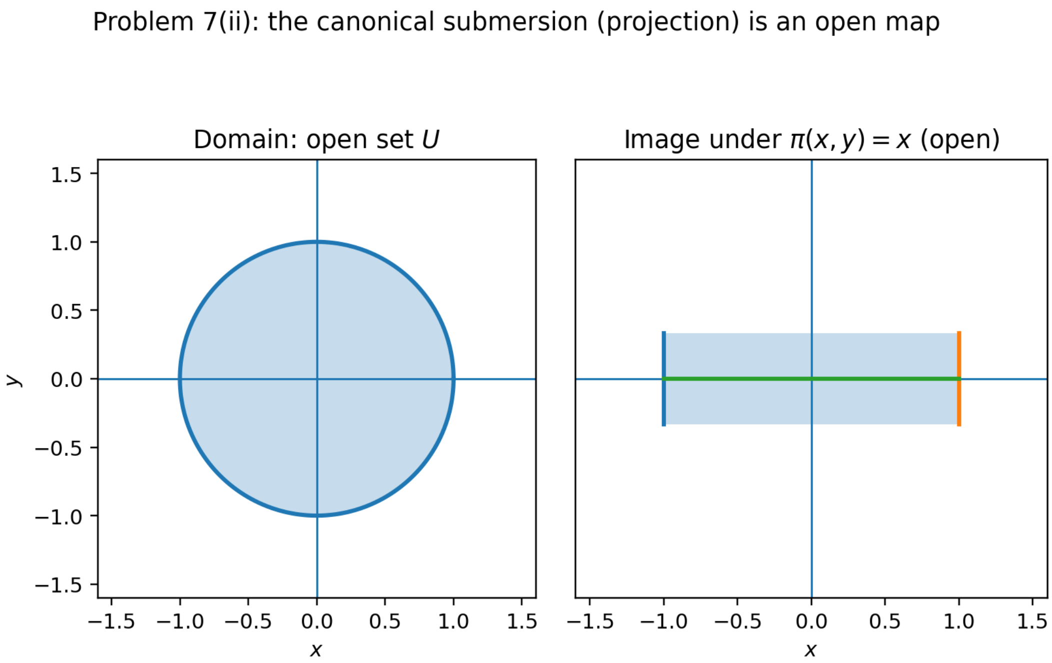


Figure 11: Problem 7(ii): the canonical submersion $\pi(x, y) = x$ is an open map: an open disk projects to an open interval.

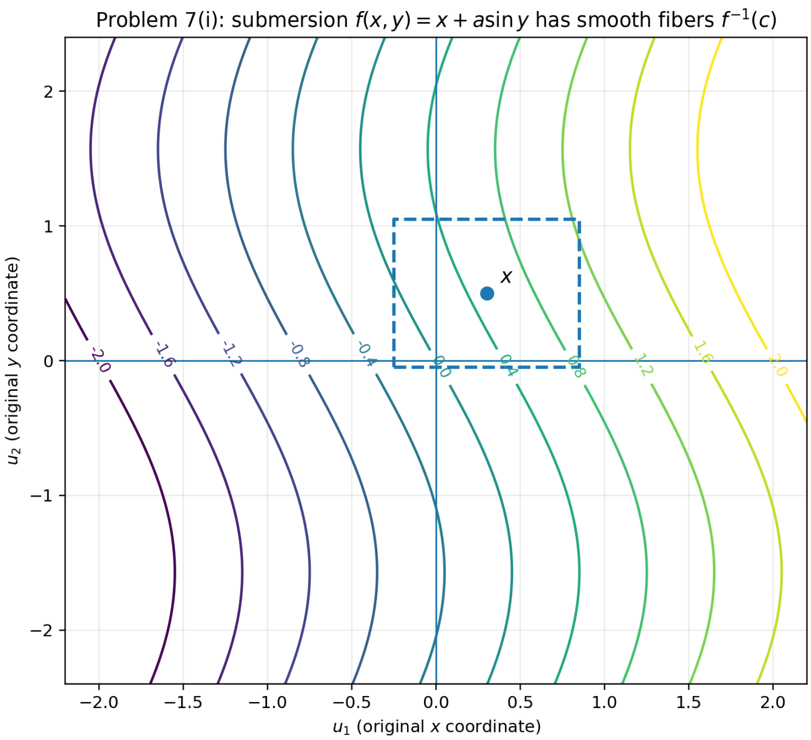


Figure 12: Problem 7(i): a nonlinear submersion $f(x, y) = x + a \sin y$ (with $a = 0.45$) has smooth fibers $f^{-1}(c)$ (level curves). The local submersion theorem says that, in suitable local coordinates, f becomes a projection and these fibers straighten to parallel coordinate slices.

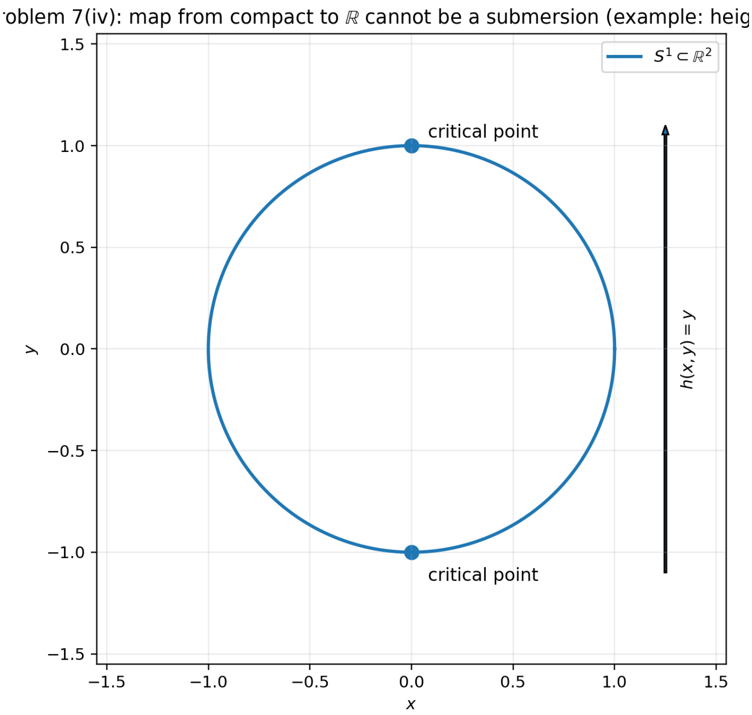


Figure 13: Problem 7(iv): typical failure on compact manifolds. The height map $h : S^1 \to \mathbb{R}$ has critical points (top/bottom), so it cannot be a submersion. In fact, no submersion from a compact manifold to $\mathbb{R}^\ell$ exists for $\ell \geq 1$.

Problem 8 (Tangent space to a regular level set)

Problem statement. Let $f : X \to Y$ be smooth and let $y \in Y$ be a regular value of f. Show that for $x \in f^{-1}(y)$,

$$T_x\big(f^{-1}(y)\big) = \ker\big(df_x : T_xX \to T_yY\big).$$

Theory / key fact. If y is a regular value, then for each $x \in f^{-1}(y)$ the differential df_x is surjective, so f is a submersion along the level set. Locally, by the local submersion theorem, f is a projection in suitable coordinates.

Solution. ($\subset$) Let $v \in T_x(f^{-1}(y))$. Then there exists a smooth curve $\gamma : (-\varepsilon, \varepsilon) \to f^{-1}(y)$ with $\gamma(0) = x$ and $\gamma'(0) = v$. Since $f(\gamma(t)) = y$ is constant,

$$0 = \frac{d}{dt}\Big|_{t=0} f(\gamma(t)) = df_x(\gamma'(0)) = df_x(v),$$

so $v \in \ker(df_x)$.

($\supset$) Conversely, let $v \in \ker(df_x)$. Since y is a regular value, f is a submersion at x. By the local submersion theorem, choose local coordinates $u = (u_1, \dots, u_k)$ near x and coordinates near y such that

$$f(u_1, \dots, u_k) = (u_1, \dots, u_\ell), \qquad \ell = \dim Y.$$

Then locally

$$f^{-1}(y) = \{u_1 = c_1, \dots, u_\ell = c_\ell\},$$

so

$$T_x(f^{-1}(y)) = \mathrm{span}\Big\{\frac{\partial}{\partial u_{\ell+1}}, \ldots, \frac{\partial}{\partial u_k}\Big\}.$$

In these coordinates df_x is the projection onto the first ℓ components, hence

$$\ker(df_x) = \mathrm{span}\Big\{\frac{\partial}{\partial u_{\ell+1}}, \ldots, \frac{\partial}{\partial u_k}\Big\} = T_x(f^{-1}(y)).$$

Therefore $T_x(f^{-1}(y)) = \ker(df_x)$. $\square$

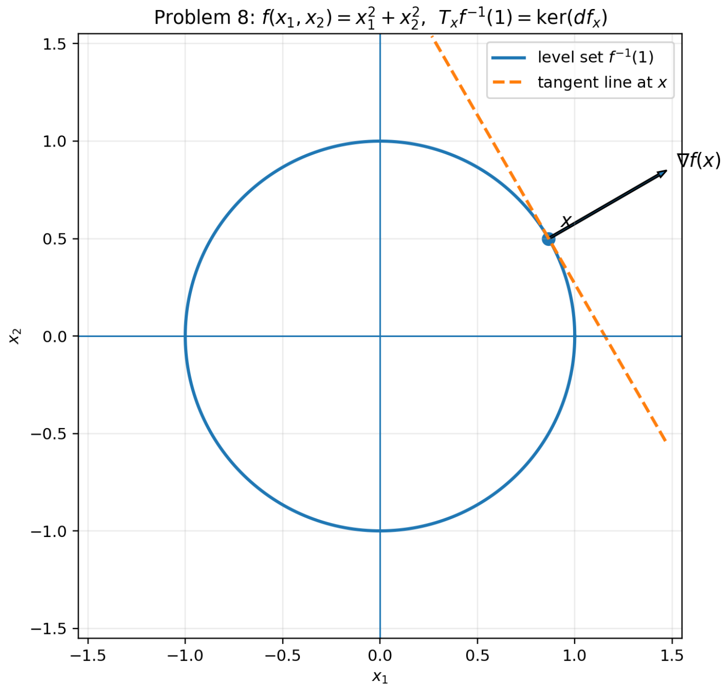


Figure 14: Problem 8: $f(x_1, x_2) = x_1^2 + x_2^2$. The level set $f^{-1}(1)$ is a circle. The tangent line at x is orthogonal to $\nabla f(x)$, and equals $\ker(df_x)$.

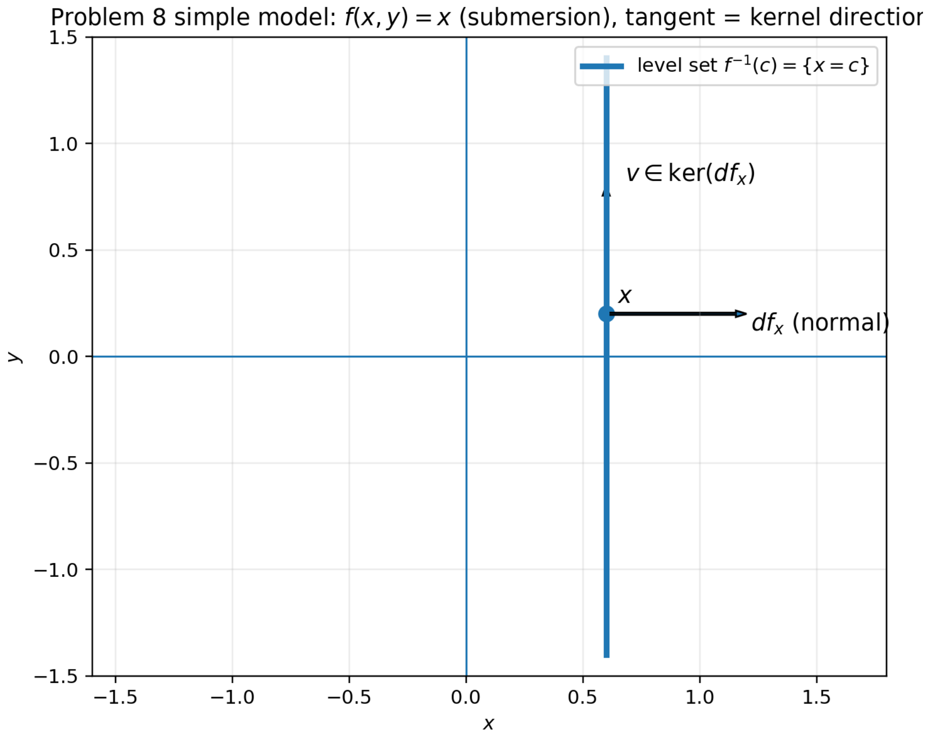


Figure 15: Problem 8 (model case): $f(x, y) = x$ is a submersion. The fiber $f^{-1}(c) = \{x = c\}$ has tangent direction $(0, 1)$, which is exactly $\ker(df_x)$.

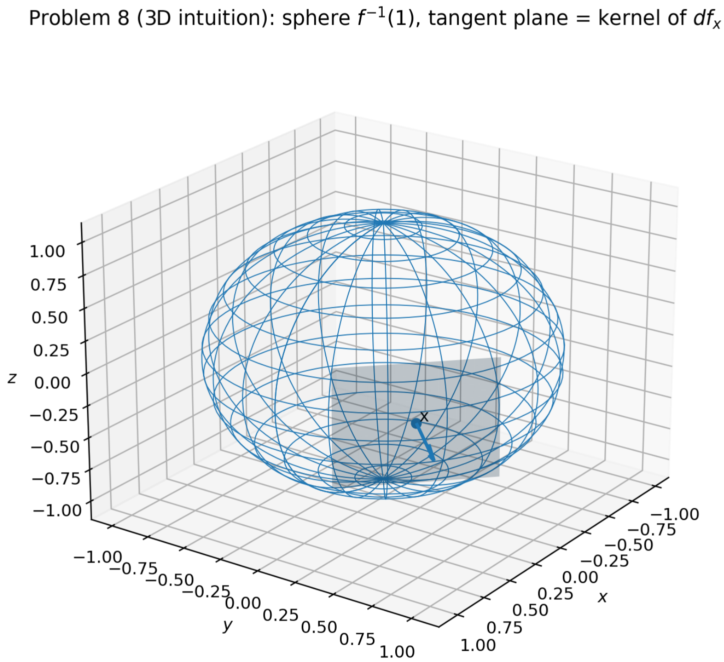


Figure 16: Problem 8 (3D intuition): for $f(x, y, z) = x^2 + y^2 + z^2$, the level set $f^{-1}(1)$ is the unit sphere. The tangent plane at x consists of vectors orthogonal to the normal direction (gradient), i.e. $\ker(df_x)$.

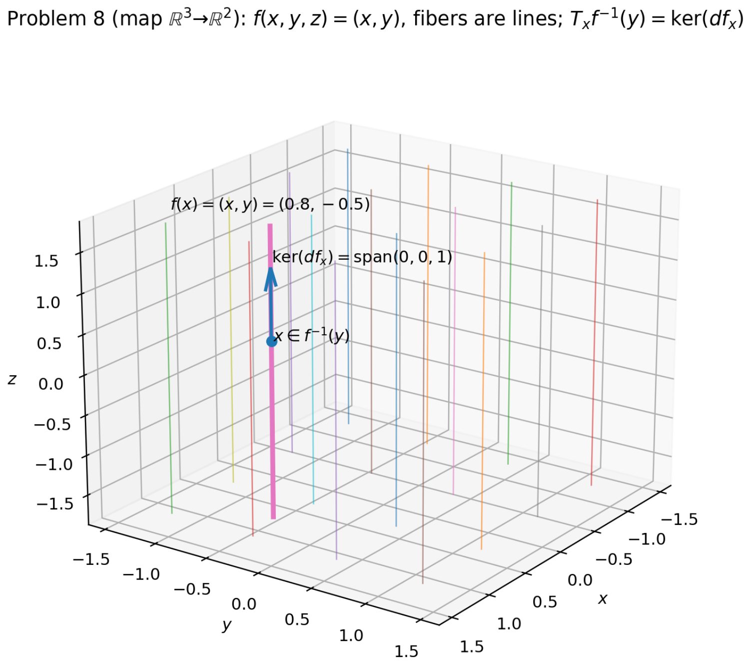


Figure 17: Problem 8 example: $f : \mathbb{R}^3 \to \mathbb{R}^2$, $f(x, y, z) = (x, y)$. For $y_0 = (a, b)$, the fiber $f^{-1}(y_0) = \{(a, b, z)\}$ is a line. Its tangent direction is the z-direction, which equals $\ker(df_x)$ since $df_x(v_x, v_y, v_z) = (v_x, v_y)$.

Problem 9 (Lefschetz map on a compact manifold has finitely many fixed points)

Statement. Let $f : X \to X$ be smooth. Assume that for every fixed point x of f, the linear map $Df_x : T_xX \to T_xX$ does not have 1 as an eigenvalue. Prove: if X is compact and f is Lefschetz, then f has only finitely many fixed points.

Solution. Define the smooth map

$$F := f - \mathrm{id}_X : X \longrightarrow X.$$

Then

$$\mathrm{Fix}(f) = \{x \in X : \ f(x) = x\} = F^{-1}(0),$$

where 0 denotes the origin in local coordinates.

Let $x \in \mathrm{Fix}(f)$. In a local chart around x we may view F as a smooth map $\widetilde{F} : \mathbb{R}^n \to \mathbb{R}^n$ with $\widetilde{F}(0) = 0$. Its derivative at 0 equals

$$D\widetilde{F}_0 \ = \ Df_x - I.$$

By hypothesis, 1 is not an eigenvalue of Df_x, hence $Df_x - I$ is invertible. Therefore $D\widetilde{F}_0$ is invertible, and by the inverse function theorem $\widetilde{F}$ is a diffeomorphism from some neighborhood of 0 onto a neighborhood of 0. In particular, the equation $\widetilde{F}(u) = 0$ has the unique solution $u = 0$ in that neighborhood. Translating back to X, we conclude that x is an isolated point of $\mathrm{Fix}(f)$.

Hence $\mathrm{Fix}(f)$ is a discrete subset of X. If $\mathrm{Fix}(f)$ were infinite, compactness of X would imply that $\mathrm{Fix}(f)$ has an accumulation point in X, contradicting discreteness. Therefore $\mathrm{Fix}(f)$ is finite.

Extra theory for Problem 9 (Lefschetz / nondegenerate fixed points)

Definition (nondegenerate fixed point). Let $f : X \to X$ be smooth and let $x \in X$ be a fixed point ($f(x) = x$). The fixed point x is called *nondegenerate* if

$$\det(I - Df_x) \neq 0,$$

equivalently, 1 is not an eigenvalue of the linear map $Df_x : T_xX \to T_xX$.

Definition (Lefschetz map, differential version). A smooth map $f : X \to X$ is called a *Lefschetz map* if *every* fixed point of f is nondegenerate, i.e. if for every x with $f(x) = x$ the map Df_x has no eigenvalue 1.

Geometric interpretation (graph vs. diagonal). Consider the smooth map

$$G : X \longrightarrow X \times X, \qquad G(p) = (p, f(p)).$$

Its image is the graph $\Gamma_f \subset X \times X$. Let

$$\Delta := \{(p, p) \in X \times X\}$$

be the diagonal. Then fixed points correspond to intersections:

$$p \in \mathrm{Fix}(f) \quad \Longleftrightarrow \quad (p, f(p)) \in \Gamma_f \cap \Delta.$$

At a fixed point x we have $(x,x) \in \Gamma_f \cap \Delta$ and the tangent spaces are

$$T_{(x,x)}\Delta = \{(v,v) : v \in T_xX\}, \qquad T_{(x,x)}\Gamma_f = \{(v, Df_xv) : v \in T_xX\}.$$

The intersection is *transverse* at (x,x) if

$$T_{(x,x)}\Gamma_f + T_{(x,x)}\Delta \;=\; T_{(x,x)}(X \times X) \cong T_xX \oplus T_xX.$$

A standard linear-algebra computation shows that this transversality condition is equivalent to $I - Df_x$ being invertible, i.e. to 1 not being an eigenvalue of Df_x. Thus:

$$f \text{ is Lefschetz} \quad \Longleftrightarrow \quad \Gamma_f \pitchfork \Delta \text{ at every point of } \Gamma_f \cap \Delta.$$

Consequences. If x is a nondegenerate fixed point, then it is *isolated*: locally the equation $f(p) = p$ has a unique solution near x. Equivalently, $\mathrm{Fix}(f)$ is a discrete subset of X. If X is compact, every discrete subset is finite, hence a Lefschetz map on a compact manifold has only finitely many fixed points.

Remark (fixed point index). If x is a nondegenerate fixed point, one defines its (local) index by

$$\mathrm{ind}(f,x) = \mathrm{sign}\,\det(I - Df_x) \in \{+1, -1\}.$$

When all fixed points are nondegenerate, the Lefschetz number can be computed as a sum of indices:

$$L(f) = \sum_{x \in \mathrm{Fix}(f)} \mathrm{ind}(f,x).$$

(For this sheet, you only need the isolation/finiteness consequence.)

Examples (Lefschetz maps and nonexamples)

Recall. A fixed point x of a smooth map $f : X \to X$ is *nondegenerate* if

$$\det(I - Df_x) \neq 0,$$

equivalently 1 is not an eigenvalue of Df_x. A map is called *Lefschetz* (in the differential sense used here) if all its fixed points are nondegenerate.

(A) Vacuous Lefschetz examples (no fixed points).

- Rotation of the circle: $f(e^{it}) = e^{i(t+\alpha)}$ with $\alpha \not\equiv 0 \pmod{2\pi}$. Then $\mathrm{Fix}(f) = \varnothing$, so f is Lefschetz.
- Translation on the torus: $T^n = \mathbb{R}^n/\mathbb{Z}^n$, $f([x]) = [x + a]$ with $a \notin \mathbb{Z}^n$. Then $\mathrm{Fix}(f) = \varnothing$.
- Antipodal map on S^n: $f(x) = -x$. Then $f(x) = x$ implies $x = 0$, impossible on S^n, so no fixed points.

(B) Lefschetz examples with isolated fixed points.

- Reflection on S^1: $f(e^{it}) = e^{-it}$. Fixed points satisfy $e^{it} = e^{-it}$, hence $t \equiv 0, \pi$. In the coordinate $t \in \mathbb{R}/2\pi\mathbb{Z}$ one has $t \mapsto -t$ so $Df = -1$ at fixed points; therefore 1 is not an eigenvalue and both fixed points are nondegenerate.

- Doubling map on $S^1 \cong \mathbb{R}/\mathbb{Z}$: $f([t]) = [2t]$. Fixed points satisfy $[2t] = [t]$, hence $t \in \mathbb{Z}$, so the unique fixed point is $[0]$. Here $Df = 2$, so $\det(I - Df) = 1 \neq 0$ and the fixed point is nondegenerate.
- Rotation of S^2 by angle $\theta \neq 0$ around the z-axis. The fixed points are the north and south poles. In the tangent plane at each pole, Df is the planar rotation by θ, whose complex eigenvalues are $e^{\pm i\theta} \neq 1$ for $\theta \not\equiv 0 \pmod{2\pi}$. Hence the fixed points are nondegenerate and the map is Lefschetz.
- Linear torus maps. Let $A \in GL(n, \mathbb{Z})$ and define $f_A : T^n \to T^n$ by $f_A([x]) = [Ax]$. Then $Df_A \equiv A$ (constant). Hence f_A is Lefschetz exactly when

  $$\det(I - A) \neq 0.$$

  In this case the fixed point set is finite; moreover one has $\#\mathrm{Fix}(f_A) = |\det(I - A)|$. Example on T^2: for $A = -I$,

  $$\det(I - (-I)) = \det(2I) = 4,$$

  and indeed $f([x]) = [-x]$ has exactly 4 fixed points (solutions of $2x \in \mathbb{Z}^2$).

(C) Nonexamples (not Lefschetz).

- The identity map id_X on any manifold: every point is fixed and $Df = I$, so 1 is an eigenvalue everywhere.
- Any map with a positive-dimensional fixed set (fixed curve/surface) is not Lefschetz. For example, reflection of S^2 across the equatorial plane fixes the entire equator S^1, so fixed points are not isolated and Df has eigenvalue 1 along tangential directions.

Example (Anosov/Lefschetz torus automorphism with explicit fixed points)

Let $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ and take

$$A = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \in SL(2, \mathbb{Z}).$$

Define the induced torus map

$$f_A : T^2 \to T^2, \qquad f_A([x]) = [Ax].$$

Hyperbolic (Anosov) property. We have $\mathrm{tr}(A) = 4$ and $\det(A) = 1$, hence the eigenvalues are

$$\lambda_\pm = \frac{4 \pm \sqrt{4^2 - 4}}{2} = 2 \pm \sqrt{3},$$

so A has one eigenvalue > 1 and one eigenvalue < 1. Therefore f_A is a hyperbolic (Anosov) torus automorphism.

Lefschetz condition. Since $Df_A \equiv A$, a fixed point is nondegenerate iff $\det(I - A) \neq 0$. Compute

$$A - I = \begin{pmatrix} 2 & 2 \\ 1 & 0 \end{pmatrix}, \qquad \det(A - I) = -2 \neq 0,$$

hence $\det(I - A) \neq 0$ and all fixed points are nondegenerate. Thus f_A is Lefschetz.

Fixed points and their number. A point $[x] \in T^2$ is fixed iff $[Ax] = [x]$, i.e.

$$(A - I)x \in \mathbb{Z}^2.$$

For linear torus maps with $\det(A - I) \neq 0$ one has

$$\#\mathrm{Fix}(f_A) = |\det(A - I)|.$$

Hence $\#\mathrm{Fix}(f_A) = 2$.

Moreover, writing $x = (x, y) \in \mathbb{R}^2$ we have

$$(A - I)(x, y) = (2x + 2y,\ x).$$

Thus $(A - I)(x, y) \in \mathbb{Z}^2$ implies $x \in \mathbb{Z}$, so $x \equiv 0 \pmod 1$, and then $2y \in \mathbb{Z}$, so $y \equiv 0$ or $1/2 \pmod 1$. Therefore the fixed points are

$$[(0,0)], \qquad \left[(0, \tfrac{1}{2})\right].$$

Problem 10 (Boundary of a manifold; the square has corners)

(i) Statement. Prove that the boundary ∂X of a smooth n-manifold with boundary X is a smooth $(n - 1)$-manifold without boundary.

Solution. By definition, for each $p \in X$ there exists a smooth chart (U, φ) with

$$\varphi : U \xrightarrow{\cong} V \subset \mathbb{H}^n, \qquad \mathbb{H}^n = \{u \in \mathbb{R}^n :\ u_n \geq 0\},$$

such that coordinate changes are smooth up to the boundary.

A point $p \in X$ lies in ∂X iff in some (hence any) boundary chart one has $\varphi(p) \in \partial\mathbb{H}^n = \{u_n = 0\}$.

Fix $p \in \partial X$ and choose a boundary chart (U, φ) around p with $\varphi(U) = V \subset \mathbb{H}^n$. Let $\pi : \mathbb{R}^n \to \mathbb{R}^{n-1}$ be the projection $\pi(u_1, \ldots, u_n) = (u_1, \ldots, u_{n-1})$ and define

$$\psi := \pi \circ \varphi\ :\ U \cap \partial X \longrightarrow \pi(V \cap \{u_n = 0\}) \subset \mathbb{R}^{n-1}.$$

Then ψ is a homeomorphism onto an open subset of $\mathbb{R}^{n-1}$ because $V \cap \{u_n = 0\}$ is open in $\{u_n = 0\} \cong \mathbb{R}^{n-1}$ and φ is a homeomorphism. If (U, φ) and (U', φ') are two boundary charts, the transition map on ∂X is

$$\psi' \circ \psi^{-1} = (\pi \circ \varphi') \circ (\pi \circ \varphi)^{-1},$$

which is the restriction (to $u_n = 0$) of the smooth map $\varphi' \circ \varphi^{-1}$ between open subsets of $\mathbb{H}^n$, hence is smooth as a map between open subsets of $\mathbb{R}^{n-1}$. Therefore these charts form a smooth atlas on ∂X, modeled on $\mathbb{R}^{n-1}$, so ∂X is a smooth $(n - 1)$-manifold *without* boundary.

(ii) Statement. Show that the square $[0, 1] \times [0, 1] \subset \mathbb{R}^2$ is not a smooth manifold with boundary (as a smooth submanifold-with-boundary of $\mathbb{R}^2$).

Solution. Assume for contradiction that $M := [0, 1] \times [0, 1]$ were a smooth 2-dimensional submanifold with boundary of $\mathbb{R}^2$. Then by part (i), ∂M would be a smooth 1-dimensional submanifold of $\mathbb{R}^2$ (a smooth embedded curve), so for every $p \in \partial M$ there exists a neighborhood $W \subset \mathbb{R}^2$ of p and a smooth diffeomorphism $\Phi : W \to \Phi(W) \subset \mathbb{R}^2$ such that

$$\Phi(M \cap W) = \mathbb{H}^2 \cap \Phi(W), \qquad \Phi(\partial M \cap W) = \{(x,0)\} \cap \Phi(W).$$

In particular, $\partial M \cap W$ must be the inverse image of a straight line segment under a diffeomorphism, hence must be a smooth embedded 1-manifold near p with a *unique* tangent direction at p.

Take the corner point $p = (0,0)$. For every sufficiently small neighborhood W of p, the set $\partial M \cap W$ is the union of two line segments:

$$(\{0\} \times [0,\varepsilon)) \ \cup \ ([0,\varepsilon) \times \{0\}),$$

meeting at a right angle at p. This set does not have a unique tangent line at p (it has two distinct tangent directions), so it cannot be a smooth embedded 1-submanifold near p. Contradiction.

Hence $[0,1] \times [0,1]$ is not a smooth manifold with boundary (it is a manifold with corners).

Extra theory for Problem 10 (Boundary of a manifold; corners)

Manifolds with boundary. A (smooth) n-manifold with boundary is a Hausdorff, second countable space X equipped with an atlas of charts

$$(U,\varphi), \qquad \varphi : U \xrightarrow{\cong} V \subset \mathbb{H}^n, \quad \mathbb{H}^n = \{u \in \mathbb{R}^n : \ u_n \geq 0\},$$

such that all transition maps are smooth (and extend smoothly across $u_n = 0$ in local coordinates).

Interior and boundary (intrinsic definition). A point $p \in X$ is an *interior point* if in some (equivalently any) boundary chart $\varphi(p)$ lies in the interior of $\mathbb{H}^n$, i.e. $(\varphi(p))_n > 0$. A point $p \in X$ is a *boundary point* if in some chart $\varphi(p)$ lies on

$$\partial\mathbb{H}^n = \{u \in \mathbb{R}^n : \ u_n = 0\}.$$

The set of boundary points is denoted ∂X, and the interior is denoted $\mathrm{int}(X)$. One has $X = \mathrm{int}(X) \,\dot\cup\, \partial X$.

Local model for the boundary. In a boundary chart (U,φ) with $\varphi(U) = V \subset \mathbb{H}^n$, the boundary part is

$$U \cap \partial X \ \cong \ V \cap \{u_n = 0\},$$

which is open in the hyperplane $\{u_n = 0\} \cong \mathbb{R}^{n-1}$. Projecting away the last coordinate

$$\pi(u_1,\ldots,u_n) = (u_1,\ldots,u_{n-1})$$

produces a coordinate chart for ∂X modeled on $\mathbb{R}^{n-1}$. This is the basic reason why ∂X is an $(n-1)$-manifold without boundary.

What fails for the square: corners. The subset $[0,1]^2 \subset \mathbb{R}^2$ has points (the four vertices) where a neighborhood in the set looks like a *quadrant*

$$\mathbb{R}^2_{\geq 0, \geq 0} = \{(u_1,u_2) : u_1 \geq 0, \ u_2 \geq 0\},$$

not like a half-plane $\mathbb{H}^2 = \{u_2 \geq 0\}$. At such a corner point, the boundary is locally the union of two smooth arcs meeting with a nonzero angle, so it is not a smooth 1-submanifold (there is no unique tangent direction). Therefore the square is not a smooth manifold with boundary; it is an example of a *manifold with corners*.

Remark (manifolds with corners). One can enlarge the definition by allowing local models
$$[0, \infty)^k \times \mathbb{R}^{n-k},$$
which yields the notion of a smooth manifold with corners. Then the square $[0, 1]^2$ is a 2-manifold with corners (with $k = 2$ at the vertices).

Examples for Problem 10 (manifolds with boundary vs corners)

Examples of manifolds with boundary.

- $[0, 1] \subset \mathbb{R}$ is a 1-manifold with boundary and $\partial[0, 1] = \{0, 1\}$.
- The closed disk $D^2 = \{(x, y) \in \mathbb{R}^2 : \ x^2 + y^2 \leq 1\}$ is a 2-manifold with boundary and $\partial D^2 = S^1$.
- More generally, the closed ball $D^n = \{x \in \mathbb{R}^n : \ \|x\| \leq 1\}$ is an n-manifold with boundary and $\partial D^n = S^{n-1}$.
- The cylinder $S^1 \times [0, 1]$ is a 2-manifold with boundary and
$$\partial(S^1 \times [0, 1]) = (S^1 \times \{0\}) \sqcup (S^1 \times \{1\}).$$
- The Möbius strip is a 2-manifold with boundary; its boundary is a single circle.

Examples of manifolds without boundary.

- S^n and T^n are compact manifolds without boundary.
- The open ball $\{x \in \mathbb{R}^n : \ \|x\| < 1\}$ is a manifold without boundary (modeled on $\mathbb{R}^n$).

Corner nonexample. The square $[0, 1]^2$ fails to be a smooth manifold with boundary because neighborhoods of corner points look like the quadrant $[0, \infty)^2$, not like the half-space $\mathbb{H}^2$. Equivalently, near a corner the boundary is the union of two line segments meeting at a nonzero angle, so it does not have a unique tangent direction.

Definition of the boundary and why $\dim(\partial X) = n - 1$

Definition (manifold with boundary). A smooth n-manifold with boundary is a Hausdorff, second countable space X equipped with an atlas of charts
$$(U, \varphi), \qquad \varphi : U \xrightarrow{\cong} V \subset \mathbb{H}^n, \qquad \mathbb{H}^n = \{u \in \mathbb{R}^n : \ u_n \geq 0\},$$
such that all transition maps are smooth (and extend smoothly up to $u_n = 0$).

Definition (interior and boundary). A point $p \in X$ is an *interior point* if for some (equivalently any) chart (U, φ) around p one has $(\varphi(p))_n > 0$. A point $p \in X$ is a *boundary* *point* if for some (equivalently any) chart (U, φ) around p one has $(\varphi(p))_n = 0$. The set of boundary points is denoted

$$\partial X := \{p \in X : \ (\varphi(p))_n = 0 \text{ in boundary coordinates}\},$$

and the set of interior points is denoted $\mathrm{int}(X) = X \setminus \partial X$.

Why this is well-defined. If (U, φ) and (U', φ') are boundary charts, then the transition map $\varphi' \circ \varphi^{-1}$ is a diffeomorphism between open subsets of $\mathbb{H}^n$. Such maps preserve the boundary hyperplane $\partial\mathbb{H}^n = \{u_n = 0\}$, hence the condition $(\varphi(p))_n = 0$ does not depend on the choice of boundary chart.

Why $\dim(\partial X) = n - 1$. Near a boundary point $p \in \partial X$, choose a boundary chart (U, φ) with $\varphi(U) = V \subset \mathbb{H}^n$. Then

$$U \cap \partial X \ \cong \ V \cap \{u_n = 0\},$$

and $V \cap \{u_n = 0\}$ is open in the hyperplane $\{u_n = 0\} \cong \mathbb{R}^{n-1}$. Thus ∂X is locally modeled on $\mathbb{R}^{n-1}$, so it is an $(n-1)$-dimensional manifold (without boundary).

Charts around a point; diffeomorphisms in the definition; counterexamples (Problem 10)

Local charts (interior vs boundary). Let X be a smooth n-manifold with boundary.

- If $p \in \mathrm{int}(X)$, then there is a chart (U, φ) with

  $$\varphi : U \to V \subset \mathbb{R}^n,$$

  so X looks locally like an open subset of $\mathbb{R}^n$ (a full ball).

- If $p \in \partial X$, then there is a chart (U, φ) with

  $$\varphi : U \to V \subset \mathbb{H}^n = \{u \in \mathbb{R}^n : \ u_n \geq 0\},$$

  so X looks locally like a half-ball. Moreover $U \cap \partial X$ corresponds to $V \cap \{u_n = 0\}$.

How diffeomorphisms enter (flattening the boundary). In the embedded setting (a submanifold-with-boundary of $\mathbb{R}^m$), the definition is often phrased as: for each $p \in X \subset \mathbb{R}^m$ there is a neighborhood W of p in $\mathbb{R}^m$ and a diffeomorphism $\Phi : W \to \Phi(W) \subset \mathbb{R}^m$ such that

$$\Phi(X \cap W) = \big(\mathbb{H}^n \times \{0\}\big) \cap \Phi(W), \qquad \Phi(\partial X \cap W) = \big(\{u_n = 0\} \times \{0\}\big) \cap \Phi(W).$$

Thus, near boundary points, a diffeomorphism straightens the boundary to a coordinate hyperplane.

Explicit flattening example (unit disk). Let $D^2 = \{(x, y) \in \mathbb{R}^2 : \ x^2 + y^2 \leq 1\}$ and $p = (1, 0) \in \partial D^2$. Near p the boundary circle is given by the smooth graph

$$x = \sqrt{1 - y^2} \qquad (y \text{ near } 0).$$

Define local coordinates (u, w) by

$$u = y, \qquad w = \sqrt{1-u^2} - x.$$

Then ∂D^2 corresponds to $w = 0$ and D^2 corresponds to $w \geq 0$. Moreover the Jacobian at p is invertible:

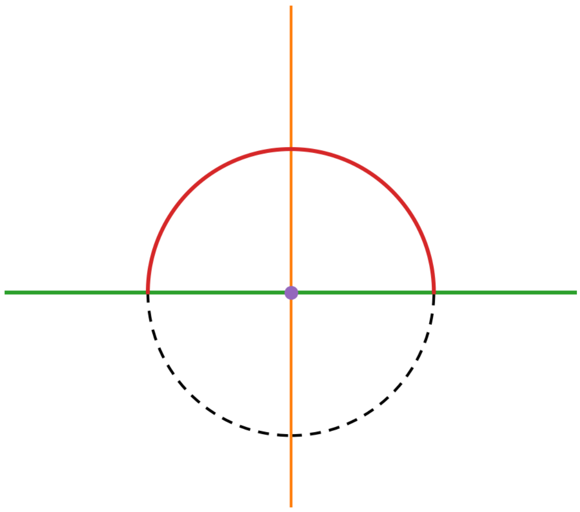
$$\frac{\partial(u, w)}{\partial(x, y)}\Big|_{(1,0)} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \det = 1 \neq 0,$$

so this is a local diffeomorphism straightening the boundary to the half-space $w \geq 0$.

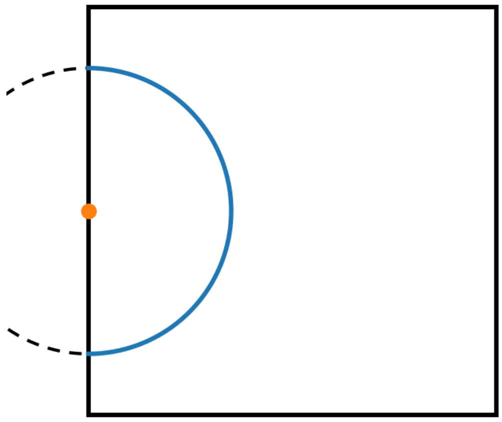
Counterexamples (why corners/cusps fail).

- The square $[0, 1]^2$ is not a smooth manifold with boundary: at a corner, neighborhoods look like a quadrant $[0, \infty)^2$, not like the half-space $\mathbb{H}^2$; equivalently the boundary has two tangent directions.
- More generally, any region with a corner (two smooth boundary arcs meeting at a nonzero angle) fails to be a manifold with boundary.
- If the boundary has a cusp (not a smooth embedded $(n-1)$-submanifold), then boundary charts with smooth transitions cannot exist.

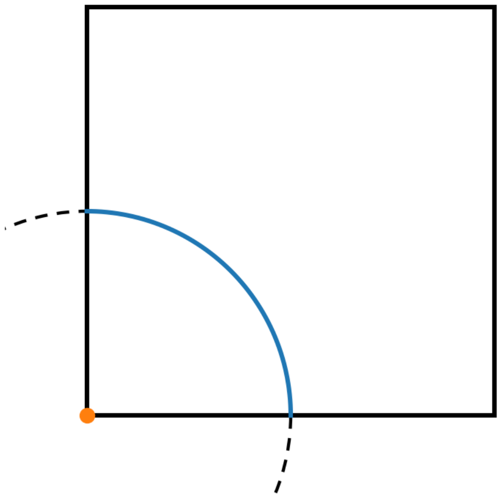
Figures.



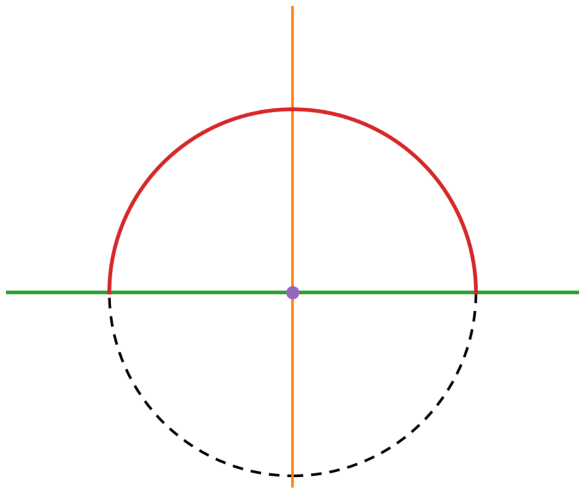
(a) Local model $\mathbb{H}^2$ with boundary $y = 0$.



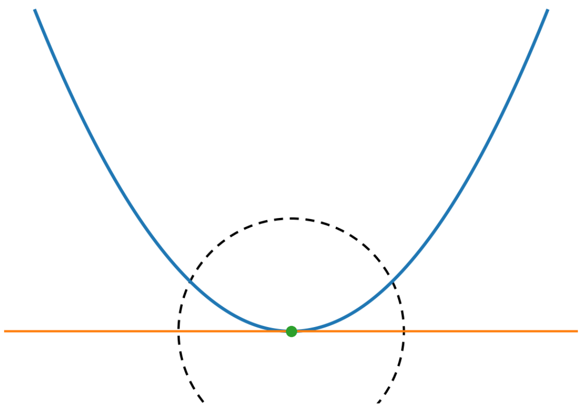
(b) Square: edge point has half-disk neighborhood.



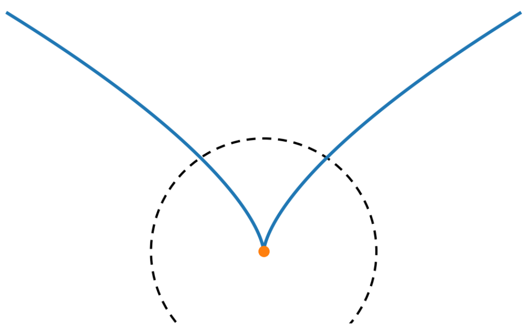
(c) Square: corner point has quarter-disk neighborhood.



(d) Flattened boundary coordinates: $w \geq 0$, boundary $w = 0$.



(e) Smooth boundary: $y = x^2$ (tangent exists).



(f) Cusp boundary: $y = |x|^{2/3}$ (not smooth at 0).

Figure 18: Neighborhood models: boundary charts look like half-spaces; corners and cusps fail.

Lemma: transition maps preserve the boundary (used in Problem 10)

Lemma. Let $U, V \subset \mathbb{H}^n$ be open and let $\psi : U \to V$ be a diffeomorphism such that ψ and ψ^{-1} extend smoothly to diffeomorphisms between open subsets of $\mathbb{R}^n$. Then

$$\psi\big(U \cap \partial\mathbb{H}^n\big) = V \cap \partial\mathbb{H}^n, \qquad \psi\big(U \cap \operatorname{int}(\mathbb{H}^n)\big) = V \cap \operatorname{int}(\mathbb{H}^n).$$

Proof. Write $\partial\mathbb{H}^n = \{u_n = 0\}$ and $\operatorname{int}(\mathbb{H}^n) = \{u_n > 0\}$. It suffices to show that ψ cannot send a boundary point to an interior point.

Assume for contradiction that $x \in U$ satisfies $x_n = 0$ but $y := \psi(x)$ satisfies $y_n > 0$. Since y is an interior point of $\mathbb{H}^n$ and V is open in $\mathbb{H}^n$, there exists a small Euclidean ball $B \subset \mathbb{R}^n$ such that

$$y \in B \subset V \subset \mathbb{H}^n \quad \text{and} \quad B \subset \{u_n > 0\}.$$

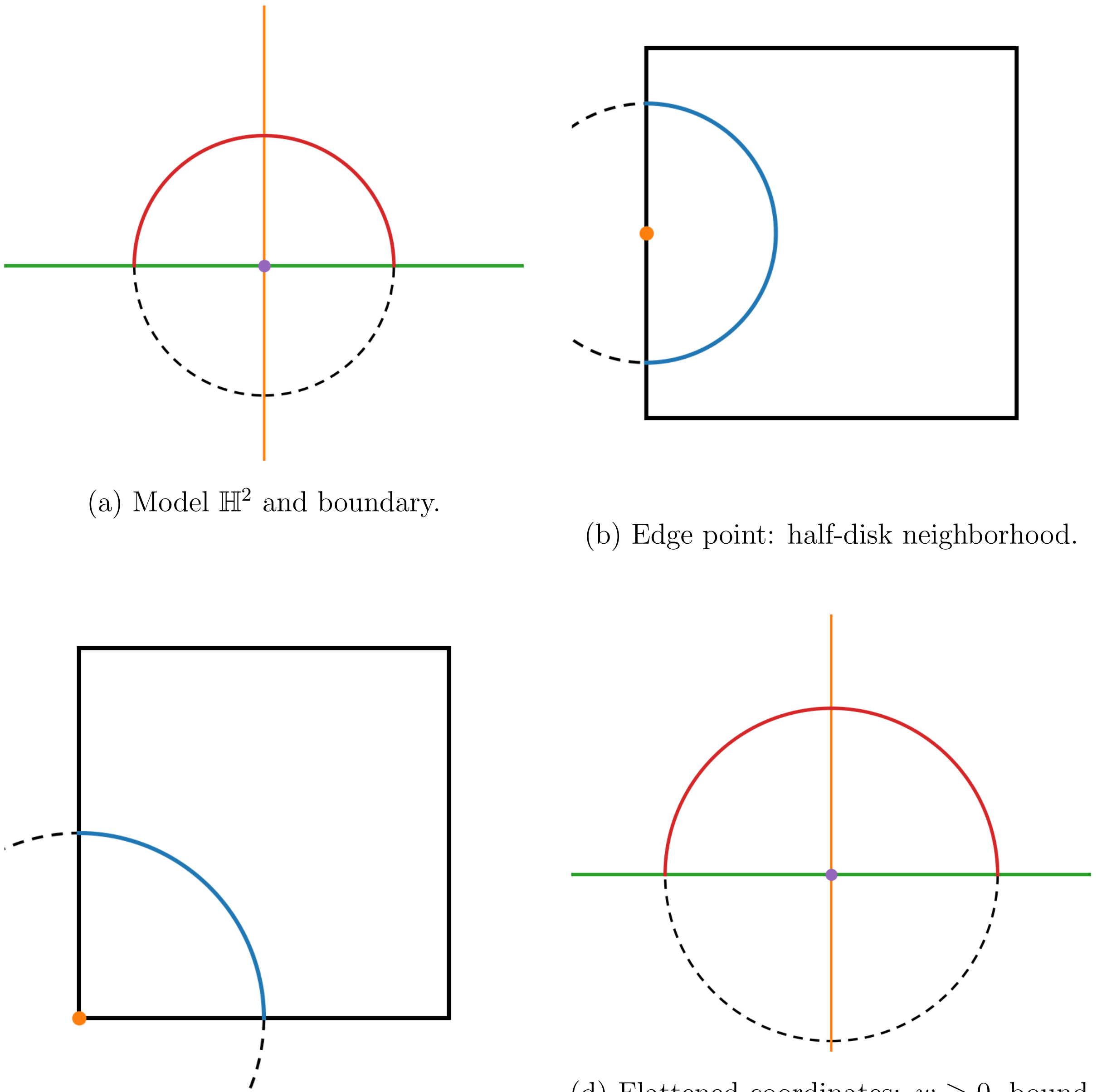
Let $\widetilde{\psi}$ be a smooth extension of ψ to an open neighborhood $W \subset \mathbb{R}^n$ of x, so that $\widetilde{\psi} : W \to \widetilde{\psi}(W)$ is a diffeomorphism and $\widetilde{\psi}|_U = \psi$. Then $\widetilde{\psi}^{-1}(B)$ is an open neighborhood of x in $\mathbb{R}^n$. In particular, it contains points with negative last coordinate, so we can choose $x' \in \widetilde{\psi}^{-1}(B)$ with $x'_n < 0$. Set $y' := \widetilde{\psi}(x') \in B$.

But $B \subset V \subset \mathbb{H}^n$, hence $y' \in V$ and $\psi^{-1}(y')$ is defined and must lie in $U \subset \mathbb{H}^n$, so its last coordinate is ≥ 0. On the other hand,

$$\psi^{-1}(y') = \widetilde{\psi}^{-1}(y') = x',$$

which has $x'_n < 0$. Contradiction.

Therefore boundary points map to boundary points, and interior points map to interior points, proving the claim. $\square$



(a) Model $\mathbb{H}^2$ and boundary.

(b) Edge point: half-disk neighborhood.

(c) Corner point: quarter-disk neighborhood.

(d) Flattened coordinates: $w \geq 0$, boundary $w = 0$.

Figure 19: Neighborhoods in the local models: boundary charts look like half-spaces; corners do not.

Problem 11 (Frobenius theorem via Brouwer fixed point)

Statement. Let A be an $n \times n$ real matrix with all entries nonnegative. Show that A has a nonnegative real eigenvalue.

Solution. If $A = 0$, then $\lambda = 0$ is a (nonnegative) eigenvalue, so assume $A \neq 0$.

If A has a zero column, say the j-th column is 0, then $Ae_j = 0$ and again $\lambda = 0$ is an eigenvalue. Hence we may assume every column of A is nonzero.

Let
$$\Delta := \Big\{x \in \mathbb{R}^n : \ x_i \ge 0 \ \forall i, \ \sum_{i=1}^n x_i = 1\Big\}.$$
This is the standard simplex; it is compact and convex.

For $x \in \Delta$ we have $Ax \ge 0$ (entrywise). Moreover $Ax \ne 0$: indeed,
$$Ax = \sum_{j=1}^n x_j \, (\mathrm{column}_j(A)),$$
is a convex combination of nonzero columns, and at least one $x_j > 0$, so $Ax \ne 0$. Therefore
$$s(x) := \sum_{i=1}^n (Ax)_i > 0 \qquad \text{for all } x \in \Delta.$$
Define
$$F : \Delta \to \Delta, \qquad F(x) := \frac{Ax}{s(x)}.$$
Then F is continuous, $F(x) \ge 0$, and
$$\sum_{i=1}^n F(x)_i = \frac{1}{s(x)} \sum_{i=1}^n (Ax)_i = 1,$$
hence $F(\Delta) \subseteq \Delta$.

By the Brouwer fixed point theorem, F has a fixed point $x \in \Delta$, i.e. $F(x) = x$. Thus
$$\frac{Ax}{s(x)} = x \quad \Longrightarrow \quad Ax = s(x)\, x.$$
Set $\lambda := s(x) \in \mathbb{R}$. Since $s(x) = \sum_i (Ax)_i$ and $Ax \ge 0$, we have $\lambda \ge 0$. Therefore $Ax = \lambda x$ with $\lambda \ge 0$, i.e. λ is a nonnegative real eigenvalue of A.

Extra theory for Problem 11 (Nonnegative matrices and Perron–Frobenius ideas)

Entrywise order and positivity. For vectors $x, y \in \mathbb{R}^n$ we write $x \ge 0$ if $x_i \ge 0$ for all i, and $x > 0$ if $x_i > 0$ for all i. For a matrix $A = (a_{ij})$ we write $A \ge 0$ if $a_{ij} \ge 0$ for all i, j. Then $A \ge 0$ implies:
$$x \ge 0 \implies Ax \ge 0.$$
This is the basic monotonicity property of nonnegative matrices.

The simplex and normalization. Define the standard simplex
$$\Delta := \Big\{x \in \mathbb{R}^n : \ x_i \ge 0 \ \forall i, \ \sum_{i=1}^n x_i = 1\Big\}.$$
It is compact and convex. For $A \ge 0$ one has $Ax \ge 0$ for all $x \in \Delta$. If additionally $Ax \ne 0$ for all $x \in \Delta$ (for example, if A has no zero column), we can normalize by the ℓ^1-sum
$$s(x) := \sum_{i=1}^n (Ax)_i > 0$$

and define a self-map of the simplex

$$F(x) := \frac{Ax}{s(x)} \in \Delta.$$

This normalization turns the linear map $x \mapsto Ax$ into a map that preserves the constraint $\sum x_i = 1$.

Fixed point theorem ⇒ eigenvector. If $F(x) = x$, then

$$\frac{Ax}{s(x)} = x \quad \Longrightarrow \quad Ax = s(x)\,x.$$

Thus x is an eigenvector and $\lambda := s(x)$ is the corresponding eigenvalue. Since $Ax \geq 0$, we get $\lambda = s(x) = \sum_i (Ax)_i \geq 0$. Therefore a fixed point of F produces a nonnegative eigenvalue of A.

Why Brouwer applies. The Brouwer fixed point theorem states: every continuous map from a compact convex set in $\mathbb{R}^n$ to itself has a fixed point. Since Δ is compact and convex, any continuous $F : \Delta \to \Delta$ has a fixed point. In Problem 11, the main work is to set up a continuous normalization map F so that:

$$\text{fixed point of } F \implies \text{eigenpair of } A \text{ with } \lambda \geq 0.$$

Remark (connection to Perron–Frobenius). Problem 11 is a first step toward the Perron–Frobenius theorem. If $A \geq 0$ is *irreducible* (or *primitive*), then one can prove much more: there exists a strictly positive eigenvector $v > 0$ and an eigenvalue $\rho(A) > 0$ (the spectral radius) such that $\rho(A)$ is simple and dominates all other eigenvalues in modulus. Problem 11 only asks for existence of *some* real eigenvalue $\lambda \geq 0$.

Examples for Problem 11 (explicit computations)

Example 1. Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \geq 0.$$

Then

$$\det(A - \lambda I) = \det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1).$$

Hence the eigenvalues are $\lambda = 3$ and $\lambda = 1$, both nonnegative. For $\lambda = 3$ one may take $v = (1, 1)$, since

$$Av = (3, 3) = 3(1, 1).$$

Moreover, with $x = (\frac{1}{2}, \frac{1}{2}) \in \Delta$ one has $Ax = (\frac{3}{2}, \frac{3}{2})$ and

$$F(x) = \frac{Ax}{\sum_i (Ax)_i} = \frac{(\frac{3}{2}, \frac{3}{2})}{3} = \left(\tfrac{1}{2}, \tfrac{1}{2}\right) = x,$$

so x is a Brouwer fixed point and produces the eigenvalue 3.

Example 2. Let

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \geq 0.$$

Then
$$\det(A-\lambda I)=\det\begin{pmatrix}-\lambda & 1\\ 1 & -\lambda\end{pmatrix}=\lambda^2-1,$$
so $\lambda=1,-1$. The nonnegative eigenvalue is $\lambda=1$ with eigenvector $v=(1,1)$, since
$$Av=(1,1)=1\cdot(1,1).$$

Example 3 (zero column). Let
$$A=\begin{pmatrix}1 & 0\\ 2 & 0\end{pmatrix}\geq 0.$$
The second column is zero, hence $Ae_2=0$, so $\lambda=0$ is an eigenvalue with eigenvector $e_2=(0,1)\geq 0$. (Indeed,
$$\det(A-\lambda I)=\det\begin{pmatrix}1-\lambda & 0\\ 2 & -\lambda\end{pmatrix}=-\lambda(1-\lambda),$$
so the eigenvalues are 0 and 1.)

Additional 3×3 example for Problem 11

Example. Let
$$A=\begin{pmatrix}1 & 1 & 0\\ 0 & 1 & 1\\ 1 & 0 & 1\end{pmatrix}\geq 0.$$
Take $v=(1,1,1)^T$. Then
$$Av=\begin{pmatrix}1+1+0\\ 0+1+1\\ 1+0+1\end{pmatrix}=\begin{pmatrix}2\\ 2\\ 2\end{pmatrix}=2v.$$
Hence A has the nonnegative real eigenvalue $\lambda=2$ with a strictly positive eigenvector $v>0$.

Brouwer normalization viewpoint. Let $\Delta=\{x\in\mathbb{R}^3:\ x_i\geq 0,\ \sum_i x_i=1\}$ and define
$$F(x)=\frac{Ax}{\sum_{i=1}^3 (Ax)_i}.$$
For $x=(\frac{1}{3},\frac{1}{3},\frac{1}{3})\in\Delta$ we have
$$Ax=\tfrac{1}{3}Av=\left(\tfrac{2}{3},\tfrac{2}{3},\tfrac{2}{3}\right),\qquad \sum_i (Ax)_i=2,$$
so
$$F(x)=\frac{(\frac{2}{3},\frac{2}{3},\frac{2}{3})}{2}=\left(\tfrac{1}{3},\tfrac{1}{3},\tfrac{1}{3}\right)=x.$$
Thus x is a fixed point of F, yielding the eigenpair $Ax=\lambda x$ with $\lambda=2\geq 0$.

Problem 12 (A compact boundaryless manifold is contractible only if it is a point)

Statement. A manifold is *contractible* if id_X is homotopic to a constant map. Show that a compact manifold without boundary is contractible only if it is the one-point space.

Solution. Let X be a compact n-manifold without boundary. If X is contractible, then X has the homotopy type of a point, so its reduced homology groups vanish:

$$\widetilde{H}_k(X;\mathbb{Z}_2) = 0 \quad \text{for all } k \geq 0.$$

In particular, if $n > 0$ then $H_n(X;\mathbb{Z}_2) = 0$.

On the other hand, every compact n-manifold without boundary carries a (mod 2) fundamental class, hence has nontrivial top homology:

$$H_n(X;\mathbb{Z}_2) \cong \mathbb{Z}_2 \neq 0.$$

(This is a standard consequence of Poincaré duality / the existence of a fundamental class for closed manifolds.) This contradicts contractibility when $n > 0$. Therefore $n = 0$.

A compact 0-manifold without boundary is a finite discrete set of points. If it is contractible, it must be connected, hence consists of exactly one point. Thus the only contractible compact manifold without boundary is the one-point space.

Remark (degree mod 2, if available). If X is a compact n-manifold without boundary, then $\deg_2(\mathrm{id}_X) = 1$. A constant map $c : X \to X$ is not onto (for $n > 0$), hence $\deg_2(c) = 0$. Since $\deg_2$ is homotopy invariant, id_X cannot be homotopic to c unless $n = 0$, giving the same conclusion.

Extra theory for Problem 12 (Contractibility vs. closed manifolds)

Contractible spaces. A topological space X is *contractible* if id_X is homotopic to a constant map. Equivalently, there exists a homotopy $H : X \times [0,1] \to X$ with

$$H(x,0) = x, \qquad H(x,1) = x_0 \text{ (independent of } x).$$

Contractible spaces have the homotopy type of a point.

Homotopy invariants: reduced homology. If X is contractible, then all reduced homology groups vanish:

$$\widetilde{H}_k(X;R) = 0 \quad \text{for all } k \geq 0$$

(for any coefficient ring R), because $X \simeq \{\mathrm{pt}\}$. In particular, for $n > 0$:

$$H_n(X;\mathbb{Z}_2) = 0.$$

Closed manifolds have nontrivial top homology (mod 2). A *compact manifold without boundary* is often called a *closed* manifold. If X is a closed n-manifold, then it possesses a (mod 2) fundamental class

$$[X] \in H_n(X;\mathbb{Z}_2),$$

and in fact

$$H_n(X;\mathbb{Z}_2) \cong \mathbb{Z}_2 \neq 0.$$

This can be proved using the existence of a fundamental class and/or Poincaré duality. Working with $\mathbb{Z}_2$ coefficients avoids any orientation issues.

Main contradiction for $n > 0$. Combining the two bullets above:

$$X \text{ contractible } \implies H_n(X; \mathbb{Z}_2) = 0,$$

but

$$X \text{ closed } n\text{-manifold } \implies H_n(X; \mathbb{Z}_2) \cong \mathbb{Z}_2 \neq 0.$$

Therefore a closed manifold cannot be contractible unless $n = 0$.

What happens in dimension 0. A smooth 0-manifold without boundary is a discrete set of points. Compactness implies it is *finite*. Contractibility implies it is *connected*, hence it must consist of exactly one point.

Remark (degree / Lefschetz viewpoint, if known). On a closed n-manifold one can define degree (e.g. mod 2), which is a homotopy invariant. The identity map has degree 1, while any constant map has degree 0 (for $n > 0$), so id_X cannot be homotopic to a constant map unless $n = 0$. This gives an alternative proof of Problem 12.

Examples for Problem 12 (Contractible vs. non-contractible spaces)

Contractible spaces (examples).

- A point $\{\mathrm{pt}\}$ (trivially contractible).
- Euclidean space $\mathbb{R}^n$ (contract via $H(x,t) = (1-t)x$).
- The open ball $B^n = \{x \in \mathbb{R}^n : \|x\| < 1\}$ and the closed ball $D^n = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$ (contract to the center by straight-line homotopy).
- Any convex subset $C \subset \mathbb{R}^n$ (e.g. intervals, rectangles, simplices): for any $x_0 \in C$, the homotopy $H(x,t) = (1-t)x + tx_0$ stays in C.
- Any star-shaped subset $S \subset \mathbb{R}^n$ (contract to a star-center by line segments).
- Cones: for any space Y, the cone $CY = (Y \times [0,1])/(Y \times \{0\})$ is contractible. (Example: the cone over S^1 is a disk.)

Non-contractible spaces (examples).

- The circle S^1 (not contractible since $\pi_1(S^1) \cong \mathbb{Z}$, equivalently $H_1(S^1) \neq 0$).
- Spheres S^n for $n \geq 1$ (not contractible since $H_n(S^n) \cong \mathbb{Z} \neq 0$).
- The torus $T^2 = S^1 \times S^1$ (not contractible since $\pi_1(T^2) \cong \mathbb{Z}^2$, equivalently $H_1(T^2) \neq 0$).
- Real projective space $\mathbb{RP}^n$ for $n \geq 1$ (not contractible; e.g. $\pi_1(\mathbb{RP}^n) \cong \mathbb{Z}_2$ for $n \geq 2$).
- Any compact manifold without boundary of positive dimension (by Problem 12).

Useful contrasts (spaces that look "almost contractible" but are not).

- The punctured plane $\mathbb{R}^2 \setminus \{0\}$ deformation retracts onto S^1, hence is not contractible.
- More generally, for $n \geq 2$, $\mathbb{R}^n \setminus \{0\}$ deformation retracts onto S^{n-1}, hence is not contractible.
- The cylinder $S^1 \times [0, 1]$ deformation retracts onto S^1, hence is not contractible.

Rule of thumb. If X is contractible, then $\widetilde{H}_k(X; R) = 0$ for all k (any coefficients R). A closed (compact boundaryless) n-manifold has $H_n(X; \mathbb{Z}_2) \cong \mathbb{Z}_2 \neq 0$ for $n > 0$, so it cannot be contractible unless it is a point.

Top homology and the fundamental class (used in Problem 12)

What is "top homology"? If X is an n-dimensional space (for us: an n-manifold), then $H_k(X; R)$ is the k-th singular homology group (with coefficients in a ring R). The *top homology* means the group in degree n:

$$H_n(X; R).$$

For an n-manifold, $H_k(X; R) = 0$ for $k > n$, so H_n is the highest potentially nonzero homology group.

Closed manifolds have nonzero top homology (mod 2). Let X be a compact n-manifold without boundary (a *closed* manifold). Then there exists a distinguished nonzero class

$$[X] \in H_n(X; \mathbb{Z}_2),$$

called the *fundamental class mod* 2. In particular,

$$H_n(X; \mathbb{Z}_2) \cong \mathbb{Z}_2 \neq 0.$$

Working with $\mathbb{Z}_2$ avoids orientation issues: every closed manifold has a mod 2 fundamental class, even if it is not orientable.

Geometric meaning of the fundamental class. Informally, $[X]$ represents the idea that an n-dimensional manifold can be "covered" by n-dimensional simplices in a way compatible with the local Euclidean orientation structure. Concretely, if one triangulates X, then the sum (mod 2) of all n-simplices is an n-cycle, and its homology class is exactly $[X]$.

Why it implies non-contractibility. If X were contractible, then $X \simeq \{\text{pt}\}$, hence all reduced homology vanishes and in particular

$$H_n(X; \mathbb{Z}_2) = 0 \qquad (n > 0).$$

But for a closed n-manifold we have $H_n(X; \mathbb{Z}_2) \cong \mathbb{Z}_2 \neq 0$. Thus a closed manifold of positive dimension cannot be contractible.

Remark (Poincaré duality viewpoint). A standard way to prove $H_n(X; \mathbb{Z}_2) \cong \mathbb{Z}_2$ is Poincaré duality: for a closed n-manifold and any coefficients R (with suitable assumptions; always valid for $R = \mathbb{Z}_2$), there is an isomorphism

$$H_k(X; R) \;\cong\; H^{n-k}(X; R),$$

and in particular $H_n(X; \mathbb{Z}_2) \cong H^0(X; \mathbb{Z}_2) \cong \mathbb{Z}_2$ when X is connected.

Examples of top homology H_n (useful for Problem 12)

1) A point. A point is 0-dimensional, and

$$H_0(\{\mathrm{pt}\};\mathbb{Z}) \cong \mathbb{Z}, \qquad \widetilde{H}_0(\{\mathrm{pt}\};\mathbb{Z}) = 0.$$

2) Interval $[0,1]$ (a 1-manifold with boundary). The interval is contractible, hence

$$H_1([0,1];\mathbb{Z}) = 0.$$

This illustrates that having boundary can kill the top homology.

3) Circle S^1 (a closed 1-manifold).

$$H_1(S^1;\mathbb{Z}) \cong \mathbb{Z}, \qquad H_1(S^1;\mathbb{Z}_2) \cong \mathbb{Z}_2.$$

4) Sphere S^n (a closed n-manifold, $n \geq 1$).

$$H_n(S^n;\mathbb{Z}) \cong \mathbb{Z}, \qquad H_n(S^n;\mathbb{Z}_2) \cong \mathbb{Z}_2.$$

5) Disk D^n (a manifold with boundary, contractible).

$$H_n(D^n;\mathbb{Z}) = 0, \qquad H_n(D^n;\mathbb{Z}_2) = 0.$$

6) Torus $T^2 = S^1 \times S^1$ (a closed 2-manifold).

$$H_2(T^2;\mathbb{Z}) \cong \mathbb{Z}, \qquad H_2(T^2;\mathbb{Z}_2) \cong \mathbb{Z}_2.$$

More generally, for the n-torus $T^n = (S^1)^n$ one has

$$H_n(T^n;\mathbb{Z}) \cong \mathbb{Z}.$$

7) Real projective plane $\mathbb{RP}^2$ (closed, nonorientable). Over $\mathbb{Z}$ the top homology vanishes due to nonorientability:

$$H_2(\mathbb{RP}^2;\mathbb{Z}) = 0,$$

but over $\mathbb{Z}_2$ it is nontrivial:

$$H_2(\mathbb{RP}^2;\mathbb{Z}_2) \cong \mathbb{Z}_2.$$

This is a key reason to use $\mathbb{Z}_2$ coefficients in Problem 12: every closed manifold has a mod 2 fundamental class.

8) Disjoint union (components add). For example,

$$H_1(S^1 \sqcup S^1;\mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

Each connected component contributes its own top-dimensional fundamental class.

Rule of thumb. If X is a connected compact n-manifold without boundary, then

$$H_n(X;\mathbb{Z}_2) \cong \mathbb{Z}_2 \neq 0.$$

If X is contractible and $n > 0$, then $H_n(X;\mathbb{Z}_2) = 0$.

Exercise 13 (Degree mod 2 and surjectivity)

Problem statement (exact). Let X be a compact manifold without boundary and Y a connected manifold with the same dimension as X.

(i) Suppose that $f : X \to Y$ has $\deg_2(f) \neq 0$. Prove that f is onto.

(ii) If Y is not compact, prove that $\deg_2(f) = 0$ for all smooth maps $f : X \to Y$.

Theory (degree mod 2). Let X be a compact smooth n-manifold without boundary and let Y be a connected smooth n-manifold with $\dim X = \dim Y = n$. Let $f : X \to Y$ be smooth.

Regular values and finite fibres. A point $y \in Y$ is a *regular value* of f if for every $x \in f^{-1}(y)$ the differential

$$df_x : T_xX \longrightarrow T_yY$$

is surjective. Since $\dim X = \dim Y = n$, surjectivity is equivalent to df_x being an isomorphism; equivalently $\det(df_x) \neq 0$ in local coordinates. By Sard's theorem, regular values exist and form a dense subset of Y. If y is a regular value, then $f^{-1}(y)$ is a 0-dimensional submanifold of X, hence a discrete subset of X. Because X is compact, every discrete subset is finite, so

$$f^{-1}(y) = \{x_1, \dots, x_k\} \quad \text{for some } k < \infty.$$

Definition of $\deg_2(f)$. For any regular value $y \in Y$, define the *degree mod* 2 of f by

$$\deg_2(f) \ := \ |f^{-1}(y)| \pmod 2 \ \in \ \mathbb{Z}/2.$$

Thus $\deg_2(f) = 0$ means that $|f^{-1}(y)|$ is even, and $\deg_2(f) = 1$ means that $|f^{-1}(y)|$ is odd. (If $f^{-1}(y) = \varnothing$, then y is automatically a regular value and the above gives $\deg_2(f) = 0$.)

Why this is well-defined (independence of y). We show that $|f^{-1}(y)| \pmod 2$ does not depend on the chosen regular value y. Let $y_0, y_1 \in Y$ be regular values. Since Y is connected, choose a smooth path $\gamma : [0,1] \to Y$ with $\gamma(0) = y_0$ and $\gamma(1) = y_1$. By a small perturbation (generic choice), we may assume γ is transverse to f. Consider the subset

$$\widetilde{M} \ := \ \{(x,t) \in X \times [0,1] \ : \ f(x) = \gamma(t)\}.$$

Define $H : X \times [0,1] \to Y \times Y$ by $H(x,t) = (f(x), \gamma(t))$ and let $\Delta \subset Y \times Y$ be the diagonal. Then $\widetilde{M} = H^{-1}(\Delta)$, and transversality of γ to f implies H is transverse to Δ. Hence $\widetilde{M}$ is a smooth submanifold of $X \times [0,1]$ of dimension

$$\dim(X \times [0,1]) - \operatorname{codim}(\Delta) = (n+1) - n = 1,$$

so $\widetilde{M}$ is a compact smooth 1-manifold (compactness because $X \times [0,1]$ is compact).

Boundary computation. A point $(x,t) \in \widetilde{M}$ lies in the boundary of $X \times [0,1]$ iff $t \in \{0,1\}$, so

$$\partial\widetilde{M} \ = \ \widetilde{M} \cap \big(X \times \{0,1\}\big) \ \cong \ f^{-1}(y_0) \ \sqcup \ f^{-1}(y_1).$$

Therefore

$$|\partial\widetilde{M}| = |f^{-1}(y_0)| + |f^{-1}(y_1)|.$$

Lemma (boundary points of a compact 1-manifold come in pairs). Every compact smooth 1-manifold is a finite disjoint union of circles S^1 and closed intervals $[0,1]$. Each circle has empty boundary, and each interval has exactly two boundary points. Hence every compact 1-manifold has an even number of boundary points:

$$|\partial \widetilde{M}| \equiv 0 \pmod 2.$$

Conclusion (well-definedness). Combining the two displays above gives

$$|f^{-1}(y_0)| + |f^{-1}(y_1)| \equiv 0 \pmod 2,$$

hence

$$|f^{-1}(y_0)| \equiv |f^{-1}(y_1)| \pmod 2.$$

Thus $\deg_2(f)$ is independent of the choice of regular value y and is therefore well-defined.

Relation to the usual integer degree (oriented case). If X and Y are oriented, one can define the usual degree $\deg(f) \in \mathbb{Z}$ by

$$\deg(f) = \sum_{x \in f^{-1}(y)} \operatorname{sign}\big(\det(df_x)\big),$$

for any regular value y. Reducing this identity modulo 2 gives

$$\deg_2(f) \equiv \deg(f) \pmod 2,$$

since $\operatorname{sign}(\det(df_x)) \in \{\pm 1\} \equiv 1 \pmod 2$. In particular, $\deg_2$ is the "orientation-free" version of degree: it is defined without choosing orientations.

Proof.

(i) Assume for contradiction that f is not onto. Then there exists $y \in Y \setminus f(X)$. For this y we have $f^{-1}(y) = \varnothing$. In particular y is a regular value (vacuously), and therefore

$$\deg_2(f) \equiv |f^{-1}(y)| \equiv 0 \pmod 2,$$

contradicting the assumption $\deg_2(f) \neq 0$. Hence f must be onto.

(ii) Since X is compact and f is continuous, the image $f(X) \subset Y$ is compact. If Y is not compact then $f(X) \neq Y$, so choose $y \in Y \setminus f(X)$. Then $f^{-1}(y) = \varnothing$, hence (as above) y is a regular value and

$$\deg_2(f) \equiv |f^{-1}(y)| \equiv 0 \pmod 2.$$

Thus $\deg_2(f) = 0$ for every smooth map $f : X \to Y$ whenever Y is non-compact.

Examples.

- The antipodal map $A : S^n \to S^n$, $A(x) = -x$, satisfies $|A^{-1}(y)| = 1$ for every y, hence $\deg_2(A) = 1$, so A is onto.
- If Y is non-compact and X is compact, then every smooth $f : X \to Y$ has $\deg_2(f) = 0$ by part **(ii)**.

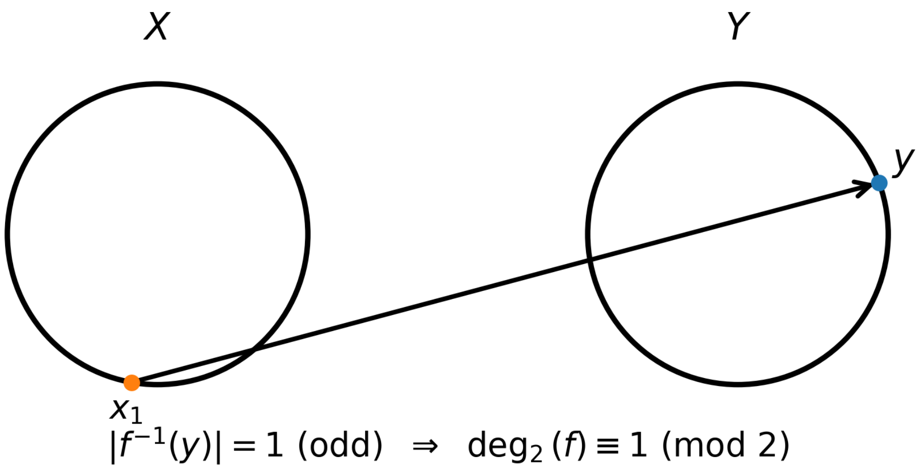


Figure 20: Odd fibre: $|f^{-1}(y)| \equiv 1 \pmod 2$ (intuition for surjectivity when $\deg_2(f) \neq 0$).

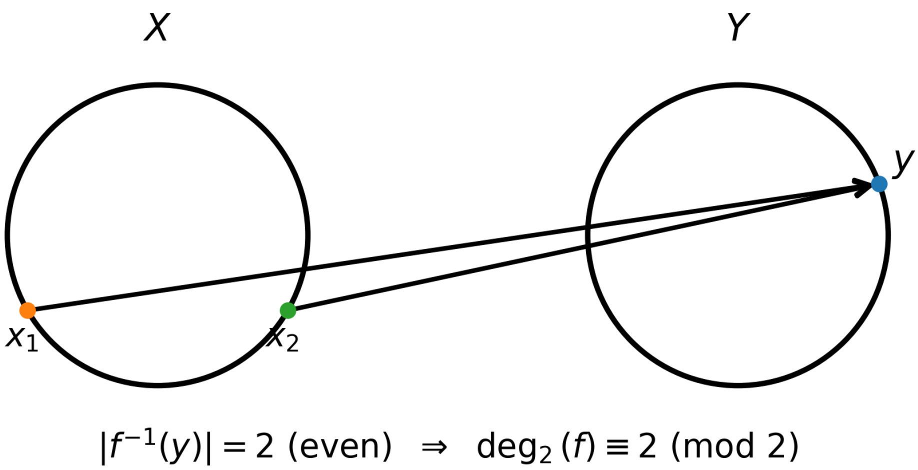


Figure 21: Even fibre: $|f^{-1}(y)| \equiv 0 \pmod 2$ (this alone does not force non-surjectivity).

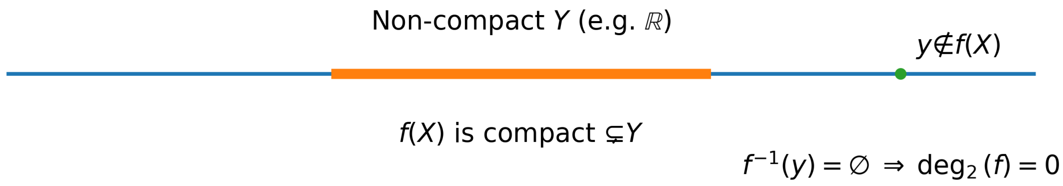


Figure 22: Non-compact target: $f(X)$ is compact, so there exists $y \notin f(X)$ and hence $\deg_2(f) = 0$.

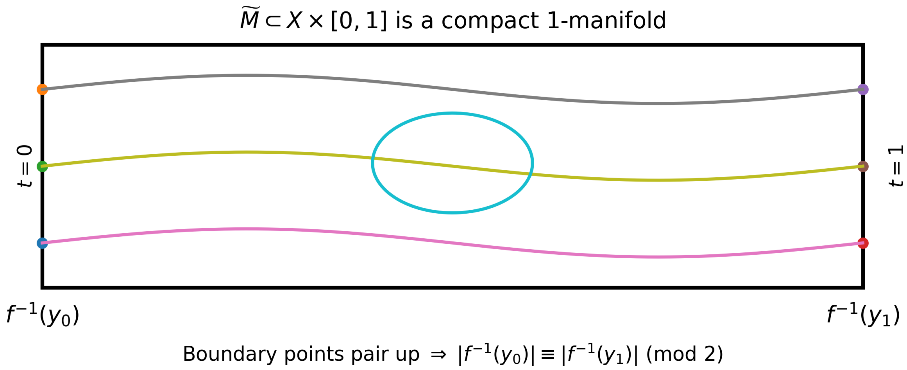


Figure 23: Well-definedness: boundary points of the preimage 1-manifold pair up, so the parity of $|f^{-1}(y)|$ is independent of y.

Figures (Exercise 13).

Exercise 14 (Bump functions and smooth glueings)

Problem statement (exact). **(Bump functions.)**

1. Let $\lambda : \mathbb{R} \to \mathbb{R}$ be given by $\lambda(x) = e^{-1/x^2}$ for $x > 0$ and $\lambda(x) = 0$ for $x \leq 0$. You know from Analysis I that λ is smooth. Show that $\tau(x) = \lambda(x-a)\lambda(b-x)$ is a smooth function, positive on (a, b) and zero elsewhere $(a < b)$.

2. Show that
$$\varphi(x) := \frac{\int_{-\infty}^{x} \tau}{\int_{-\infty}^{\infty} \tau}$$
is smooth, $\varphi(x) = 0$ for $x < a$, $\varphi(x) = 1$ for $x > b$ and $0 < \varphi(x) < 1$ for $x \in (a, b)$.

3. Finally construct a smooth function on $\mathbb{R}^n$ that equals 1 on the ball of radius a, zero outside the ball of radius b, and is strictly in between at intermediate points (here $0 < a < b$).

These functions are very useful for smooth glueings. As an illustration, suppose $f_0, f_1 : X \to Y$ are smooth homotopic maps. Show that there exists a smooth homotopy $\widetilde{F} : X \times [0,1] \to Y$ such that $\widetilde{F}(x,t) = f_0(x)$ for all $t \in [0, 1/4]$ and $\widetilde{F}(x,t) = f_1(x)$ for all $t \in [3/4, 1]$. Conclude that smooth homotopy is an equivalence relation.

(i) Since λ is smooth, the functions $x \mapsto \lambda(x-a)$ and $x \mapsto \lambda(b-x)$ are smooth (composition with translations and $x \mapsto b - x$). Hence their product

$$\tau(x) = \lambda(x-a)\lambda(b-x)$$

is smooth. If $x \leq a$ then $x - a \leq 0$ so $\lambda(x-a) = 0$, hence $\tau(x) = 0$. If $x \geq b$ then $b - x \leq 0$ so $\lambda(b-x) = 0$, hence $\tau(x) = 0$. If $a < x < b$, then $x - a > 0$ and $b - x > 0$, so both factors are strictly positive and therefore $\tau(x) > 0$ on (a, b).

(ii) Define

$$\Phi(x) := \int_{-\infty}^{x} \tau(s)\, ds, \qquad C := \int_{-\infty}^{\infty} \tau(s)\, ds.$$

Because τ is smooth and compactly supported, the Fundamental Theorem of Calculus gives $\Phi'(x) = \tau(x)$, and inductively $\Phi^{(k)}(x) = \tau^{(k-1)}(x)$ for all $k \geq 1$. Thus $\Phi \in C^\infty(\mathbb{R})$. Moreover $C > 0$ since $\tau > 0$ on (a, b), so

$$\varphi(x) = \frac{\Phi(x)}{C}$$

is smooth. If $x < a$ then $\tau(s) = 0$ for all $s \leq x$, hence $\Phi(x) = 0$ and $\varphi(x) = 0$. If $x > b$, then $\Phi(x) = C$ and $\varphi(x) = 1$. If $a < x < b$, then $0 < \Phi(x) < C$, hence $0 < \varphi(x) < 1$.

(iii) Apply (ii) with parameters $a^2 < b^2$ to obtain a smooth $\rho : \mathbb{R} \to [0, 1]$ such that $\rho(t) = 0$ for $t < a^2$ and $\rho(t) = 1$ for $t > b^2$. Define

$$\beta : \mathbb{R}^n \to \mathbb{R}, \qquad \beta(x) := 1 - \rho(\|x\|^2).$$

Then β is smooth (composition of smooth maps). If $\|x\| \leq a$ then $\|x\|^2 \leq a^2$ so $\rho(\|x\|^2) = 0$ and $\beta(x) = 1$. If $\|x\| \geq b$ then $\|x\|^2 \geq b^2$ so $\rho(\|x\|^2) = 1$ and $\beta(x) = 0$. For $a < \|x\| < b$ we have $0 < \rho(\|x\|^2) < 1$, hence $0 < \beta(x) < 1$.

Endpoint-constant smooth homotopy. Let $F : X \times [0,1] \to Y$ be a smooth homotopy with $F(\cdot, 0) = f_0$ and $F(\cdot, 1) = f_1$. Using (ii) with $a = \frac{1}{4}$ and $b = \frac{3}{4}$, construct a smooth function $\sigma : [0,1] \to [0,1]$ such that $\sigma(t) = 0$ for $t \in [0, 1/4]$ and $\sigma(t) = 1$ for $t \in [3/4, 1]$. Define

$$\widetilde{F}(x,t) := F(x, \sigma(t)).$$

Then $\widetilde{F}$ is smooth and satisfies $\widetilde{F}(x,t) = f_0(x)$ on $[0, 1/4]$ and $\widetilde{F}(x,t) = f_1(x)$ on $[3/4, 1]$.

Smooth homotopy is an equivalence relation. Reflexivity holds via the constant homotopy $H(x,t) = f(x)$. Symmetry holds by reversing time: if F is a homotopy $f_0 \sim f_1$, then $F^{\text{rev}}(x,t) := F(x, 1-t)$ is a homotopy $f_1 \sim f_0$. For transitivity, given $f_0 \sim f_1$ via F and $f_1 \sim f_2$ via G, first replace F, G by endpoint-constant homotopies as above (so F is constant near $t = 1$ and G constant near $t = 0$), and then glue:

$$H(x,t) = \begin{cases} F(x, 2t), & t \in [0, 1/2], \\ G(x, 2t-1), & t \in [1/2, 1]. \end{cases}$$

The endpoint-constant property guarantees smoothness at $t = 1/2$, so H is a smooth homotopy from f_0 to f_2.

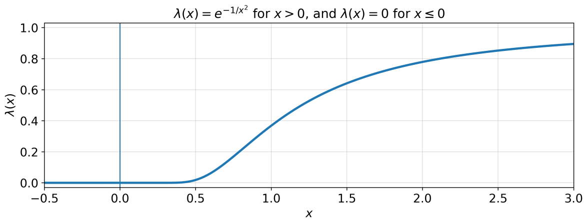


Figure 24: The basic smooth "seed" bump λ.

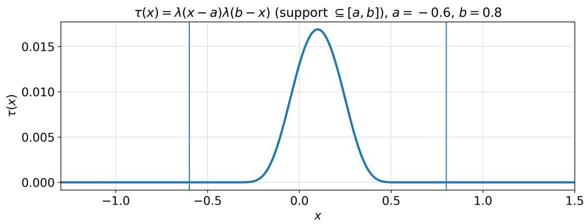


Figure 25: $\tau(x) = \lambda(x-a)\lambda(b-x)$ supported in $[a, b]$.

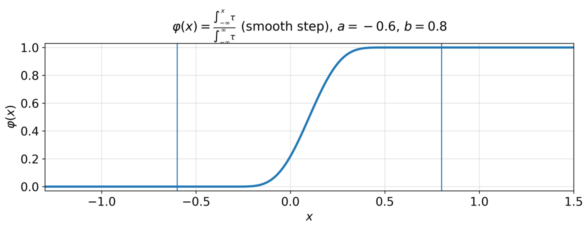


Figure 26: Smooth step φ obtained by normalizing $\int \tau$.

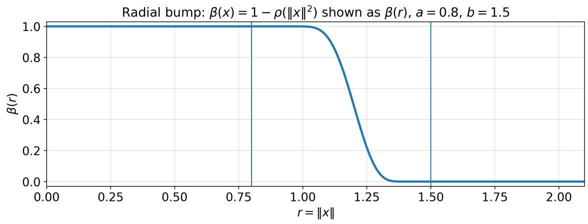


Figure 27: Radial bump on $\mathbb{R}^n$: $\beta(x) = 1 - \rho(\|x\|^2)$.

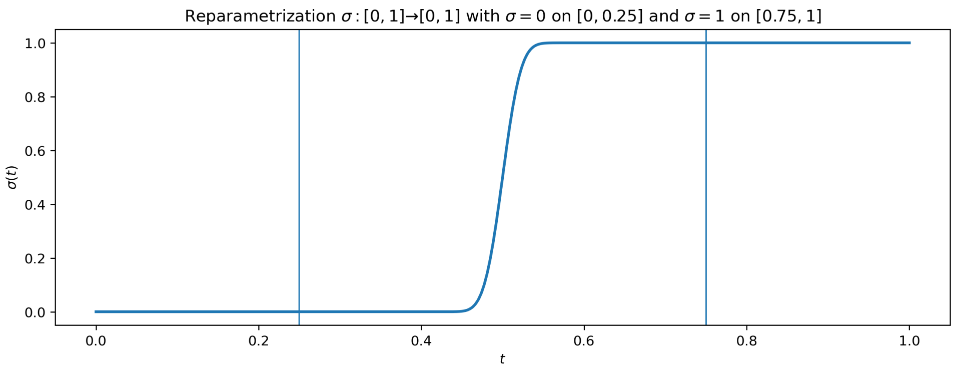


Figure 28: Endpoint-constant reparametrization σ used to flatten a homotopy near $t = 0, 1$.

Figures (Exercise 14).

Exercise 14 (Bump functions and smooth glueings)

Problem statement (exact). (Bump functions.)

1. Let $\lambda : \mathbb{R} \to \mathbb{R}$ be given by $\lambda(x) = e^{-1/x^2}$ for $x > 0$ and $\lambda(x) = 0$ for $x \leq 0$. You know from Analysis I that λ is smooth. Show that $\tau(x) = \lambda(x-a)\lambda(b-x)$ is a smooth function, positive on (a, b) and zero elsewhere ($a < b$).

2. Show that
$$\varphi(x) := \frac{\int_{-\infty}^{x} \tau}{\int_{-\infty}^{\infty} \tau}$$
is smooth, $\varphi(x) = 0$ for $x < a$, $\varphi(x) = 1$ for $x > b$ and $0 < \varphi(x) < 1$ for $x \in (a, b)$.

3. Finally construct a smooth function on $\mathbb{R}^n$ that equals 1 on the ball of radius a, zero outside the ball of radius b, and is strictly in between at intermediate points (here $0 < a < b$).

These functions are very useful for smooth glueings. As an illustration, suppose $f_0, f_1 : X \to Y$ are smooth homotopic maps. Show that there exists a smooth homotopy $\widetilde{F} : X \times [0,1] \to Y$ such that $\widetilde{F}(x,t) = f_0(x)$ for all $t \in [0, 1/4]$ and $\widetilde{F}(x,t) = f_1(x)$ for all $t \in [3/4, 1]$. Conclude that smooth homotopy is an equivalence relation.

Theory (Bump functions)

Bump functions. A *bump function* is a smooth function which is identically 0 outside a prescribed region and strictly positive (or equal to 1) inside, with a smooth transition on a boundary layer. Bump functions are the basic tool for smooth cutoffs, partitions of unity, and smooth glueing constructions.

The standard seed bump. Define

$$\lambda(x) = \begin{cases} e^{-1/x^2}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

Then $\lambda \in C^\infty(\mathbb{R})$, $\lambda(x) > 0$ for $x > 0$, and $\lambda(x) = 0$ for $x \leq 0$. Moreover λ is *flat at* 0: for every $k \geq 0$,

$$\lambda^{(k)}(0) = 0.$$

(Equivalently, e^{-1/x^2} tends to 0 as $x \to 0^+$ faster than any power of x.)

Compactly supported bump on an interval. Given $a < b$, the function

$$\tau(x) = \lambda(x - a)\lambda(b - x)$$

is smooth (composition and product of smooth functions), satisfies $\tau(x) = 0$ for $x \leq a$ and $x \geq b$, and $\tau(x) > 0$ for $a < x < b$. Hence τ is a smooth nonnegative function supported in $[a, b]$.

Smooth step function from τ. Define

$$\Phi(x) = \int_{-\infty}^{x} \tau(s)\, ds, \qquad C = \int_{-\infty}^{\infty} \tau(s)\, ds > 0, \qquad \varphi(x) = \frac{\Phi(x)}{C}.$$

By the Fundamental Theorem of Calculus, $\Phi'(x) = \tau(x)$, and iterating differentiation gives $\Phi \in C^\infty(\mathbb{R})$, hence $\varphi \in C^\infty(\mathbb{R})$. Since τ is supported in $[a, b]$, one has:

$$\varphi(x) = 0 \text{ for } x \leq a, \qquad \varphi(x) = 1 \text{ for } x \geq b, \qquad 0 < \varphi(x) < 1 \text{ for } a < x < b.$$

Radial bump on $\mathbb{R}^n$. Apply the previous construction to obtain a smooth $\rho : \mathbb{R} \to [0, 1]$ such that

$$\rho(t) = 0 \text{ for } t \leq a^2, \qquad \rho(t) = 1 \text{ for } t \geq b^2.$$

Define

$$\beta : \mathbb{R}^n \to \mathbb{R}, \qquad \beta(x) = 1 - \rho(\|x\|^2).$$

Then $\beta \in C^\infty(\mathbb{R}^n)$, $\beta(x) = 1$ for $\|x\| \leq a$, $\beta(x) = 0$ for $\|x\| \geq b$, and $0 < \beta(x) < 1$ for $a < \|x\| < b$.

Endpoint-constant reparametrization (for smooth glueing). Using the smooth step construction on $[0,1]$, choose $\sigma : [0,1] \to [0,1]$ smooth such that

$$\sigma(t) = 0 \text{ for } t \in [0, 1/4], \qquad \sigma(t) = 1 \text{ for } t \in [3/4, 1].$$

Given a smooth homotopy $F : X \times [0,1] \to Y$ from f_0 to f_1, the reparametrized homotopy

$$\widetilde{F}(x,t) = F(x, \sigma(t))$$

is smooth and satisfies $\widetilde{F}(x,t) = f_0(x)$ for $t \in [0,1/4]$ and $\widetilde{F}(x,t) = f_1(x)$ for $t \in [3/4,1]$. This endpoint-constant property is the standard tool enabling smooth glueing of homotopies and thus transitivity of smooth homotopy.

Problem 15 (Morse functions)

Problem. Let X be a k-manifold and $f : X \to \mathbb{R}$ a smooth function. Recall that $x \in X$ is a *critical point* if df_x is not surjective, i.e. $df_x = 0$. A critical point is *non-degenerate* if, in local coordinates near x, the Hessian matrix $\left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)$ has non-vanishing determinant. If all critical points of f are non-degenerate, f is called a *Morse function.*

1. Show that the condition $\det\left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right) \neq 0$ is independent of the choice of chart.
2. Suppose now that X is an open subset of $\mathbb{R}^k$. Given $a \in \mathbb{R}^k$, set

   $$f_a(x) = f(x) + \langle x, a \rangle,$$

   where $\langle x, a \rangle$ is the usual inner product. Show that f_a is a Morse function for a dense set of values of a. *(Hint: consider $\nabla f : X \to \mathbb{R}^k$.)*
3. Show that the determinant function on $M(n)$ is Morse if $n = 2$, but not if $n > 2$.

Theory

T1. Hessian under change of coordinates at a critical point. Let f be smooth and p a critical point. If $y = y(x)$ is a coordinate change with Jacobian $J = \left(\frac{\partial x_i}{\partial y_\alpha}\right)$, then at p,

$$\mathrm{Hess}_y(f) = J^T \,\mathrm{Hess}_x(f)\, J, \qquad \det(\mathrm{Hess}_y(f)) = \det(J)^2 \,\det(\mathrm{Hess}_x(f)).$$

T2. Regular values and Sard's theorem (used for generic Morse). For a smooth map $F : X \to \mathbb{R}^k$, a value v is *regular* if dF_x is surjective for all $x \in F^{-1}(v)$. Sard's theorem implies that the set of regular values of F is dense in $\mathbb{R}^k$.

T3. Differential of det **and critical matrices.** For $A \in M(n)$ and $H \in M(n)$,

$$d(\det)_A(H) = \mathrm{tr}(\mathrm{adj}(A)^T H).$$

Hence A is critical for det iff $\mathrm{adj}(A) = 0$, equivalently $\mathrm{rank}(A) \leq n-2$.

Solution

(i) Let $(x_1, \ldots, x_k)$ and $(y_1, \ldots, y_k)$ be two charts near $p \in X$ and write $F(y) = f(x(y))$. By the chain rule,

$$\frac{\partial F}{\partial y_\alpha} = \sum_i \frac{\partial f}{\partial x_i} \frac{\partial x_i}{\partial y_\alpha}.$$

Differentiating again,

$$\frac{\partial^2 F}{\partial y_\alpha \partial y_\beta} = \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j} \frac{\partial x_i}{\partial y_\alpha} \frac{\partial x_j}{\partial y_\beta} + \sum_i \frac{\partial f}{\partial x_i} \frac{\partial^2 x_i}{\partial y_\alpha \partial y_\beta}.$$

At a critical point p, $\frac{\partial f}{\partial x_i}(p) = 0$, so the second sum vanishes at p. Thus $\mathrm{Hess}_y(f) = J^T \mathrm{Hess}_x(f) J$ at p, and

$$\det(\mathrm{Hess}_y(f))(p) = \det(J(p))^2 \det(\mathrm{Hess}_x(f))(p).$$

Since $\det(J(p)) \neq 0$, we have $\det(\mathrm{Hess}_x(f))(p) \neq 0$ iff $\det(\mathrm{Hess}_y(f))(p) \neq 0$.

(ii) Let $X \subset \mathbb{R}^k$ be open and define $f_a(x) = f(x) + \langle x, a \rangle$. Then

$$\nabla f_a(x) = \nabla f(x) + a,$$

so x is a critical point of f_a iff $\nabla f(x) = -a$. Moreover $\mathrm{Hess}(f_a) = \mathrm{Hess}(f)$ because $\langle x, a \rangle$ is linear. But $d(\nabla f)_x$ is represented by $\mathrm{Hess}(f)(x)$, hence x is a non-degenerate critical point of f_a iff $-a$ is a regular value of ∇f at x. By Sard's theorem, the set of regular values of ∇f is dense in $\mathbb{R}^k$, so for a dense set of a the function f_a is Morse.

(iii) Consider $\det : M(n) \to \mathbb{R}$. By **T3**, the critical points are exactly the matrices with $\mathrm{rank}(A) \leq n - 2$.

Case $n = 2$. Write $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, so $f(a, b, c, d) = ad - bc$. The only critical point is $A = 0$. The Hessian is constant:

$$\mathrm{Hess}(f) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad \det(\mathrm{Hess}(f)) = 1 \neq 0.$$

Hence det is Morse on $M(2)$.

Case $n > 2$. The critical set contains the smooth stratum $\{A : \mathrm{rank}(A) = n - 2\}$, which has positive dimension. On this set we have $\det \equiv 0$. Therefore, for any curve $\gamma(t)$ lying in this stratum with $\gamma(0) = A$ and $\gamma'(0) = \xi \neq 0$, the function $\det(\gamma(t))$ is identically 0, so $\frac{d^2}{dt^2}\big|_{t=0} \det(\gamma(t)) = 0$. Thus the Hessian quadratic form at A vanishes on a nonzero tangent vector ξ, so the Hessian is degenerate. Hence det is not Morse for $n > 2$.

Examples and remarks

E1 (Quick check in $\mathbb{R}^k$). For $f(x) = \|x\|^2$ we have $\nabla f(x) = 2x$ and $\mathrm{Hess}(f) = 2I$, so 0 is a non-degenerate critical point: f is Morse.

E2 (Why the perturbation is "generic"). Part (ii) shows that adding a generic linear form makes all critical points non-degenerate. With additional work (using transversality), the same phenomenon holds on any manifold: a generic smooth function is Morse.

Figures.

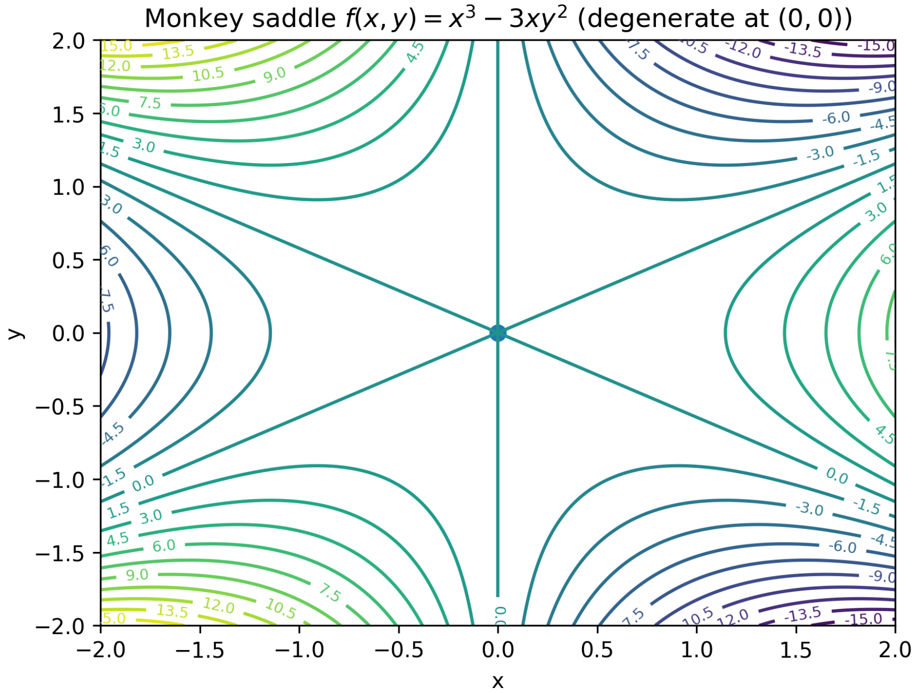


Figure 29: Example of a degenerate critical point: the monkey saddle $f(x, y) = x^3 - 3xy^2$ has a critical point at $(0, 0)$ with degenerate Hessian.

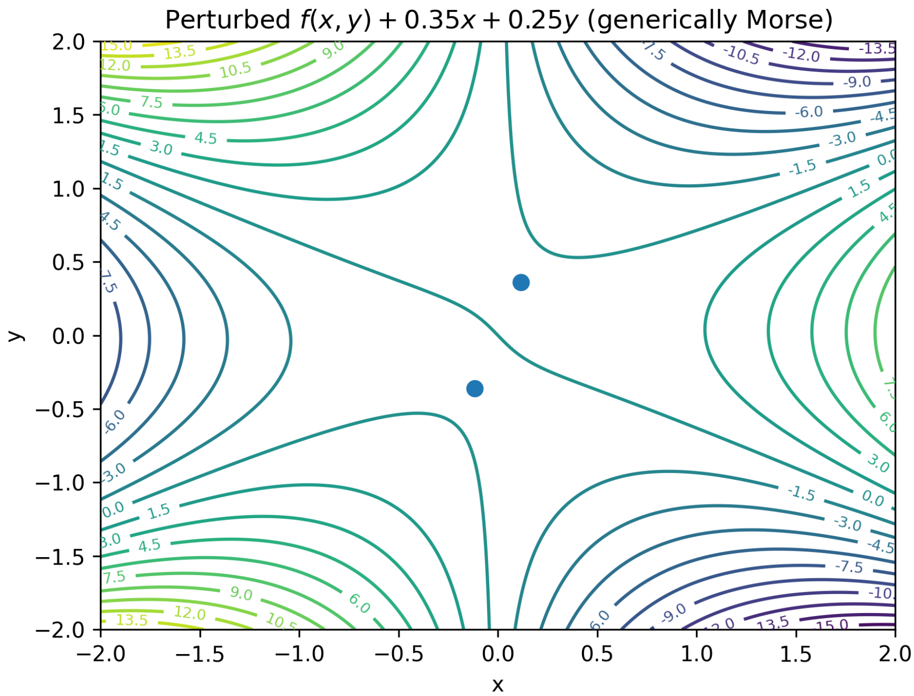


Figure 30: Generic linear perturbation $f_a = f + ax + by$ produces isolated non-degenerate critical points (Morse).

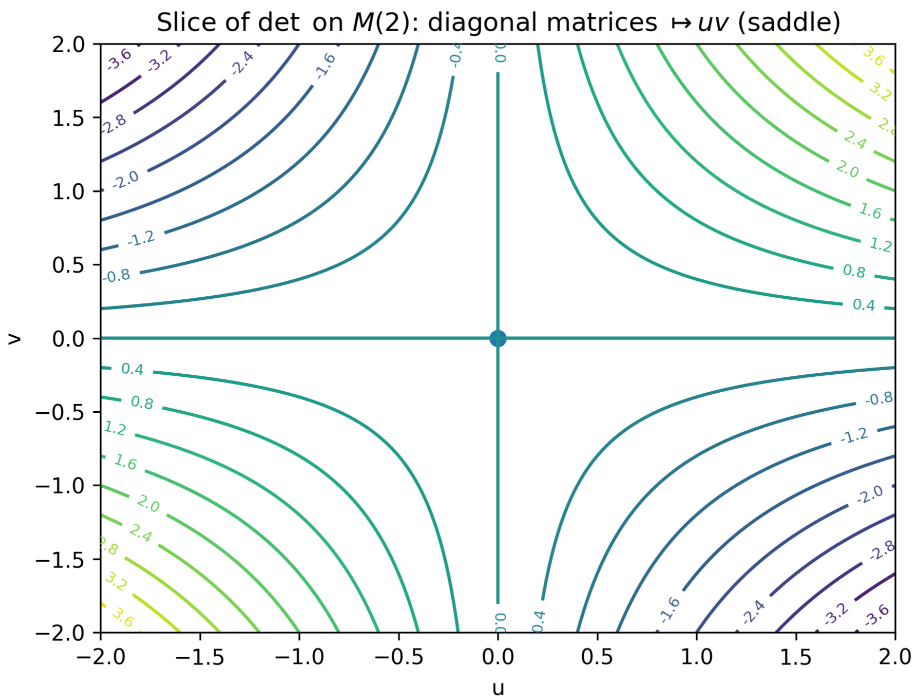


Figure 31: For $n = 2$, a diagonal slice of det is uv, which has a non-degenerate saddle critical point at $(0, 0)$.

How to recognize critical, non-degenerate, and degenerate critical points

Critical points. Let $f : \mathbb{R}^k \to \mathbb{R}$ be smooth. A point $x_0 \in \mathbb{R}^k$ is a *critical point* of f if

$$df_{x_0} = 0.$$

In standard coordinates this is equivalent to vanishing of all first partial derivatives:

$$\nabla f(x_0) = 0 \qquad \Longleftrightarrow \qquad \frac{\partial f}{\partial x_1}(x_0) = \cdots = \frac{\partial f}{\partial x_k}(x_0) = 0.$$

Geometrically, this means that the first-order change of f at x_0 is zero in every direction: there is no “instant” increase or decrease to first order.

Non-degenerate vs. degenerate. Assume x_0 is a critical point. The *Hessian matrix* of f at x_0 is

$$H_f(x_0) = \left(\frac{\partial^2 f}{\partial x_i \, \partial x_j}(x_0) \right)_{1 \le i,j \le k}.$$

- The critical point x_0 is called *non-degenerate* if

  $$\det\big(H_f(x_0)\big) \neq 0.$$

  Equivalently, $H_f(x_0)$ is invertible. In this case the quadratic (second-order) approximation of f at x_0 has genuine curvature in all directions (no “flat” direction survives).

- The critical point x_0 is called *degenerate* if

  $$\det\big(H_f(x_0)\big) = 0.$$

  Equivalently, $H_f(x_0)$ has a nontrivial kernel, so there exists a nonzero direction v such that the quadratic form $v^T H_f(x_0) v$ vanishes. In this case second-order information is not sufficient to determine the local shape, and higher-order terms in the Taylor expansion may dominate.

Example 1: a degenerate critical point (monkey saddle)

Consider

$$f(x, y) = x^3 - 3xy^2.$$

Step 1: critical point. Compute the first derivatives:

$$f_x(x, y) = 3x^2 - 3y^2, \qquad f_y(x, y) = -6xy.$$

At $(0, 0)$ we have

$$f_x(0, 0) = 0, \qquad f_y(0, 0) = 0,$$

so $(0, 0)$ is a critical point.

Step 2: degeneracy. Compute the second derivatives:

$$f_{xx}(x, y) = 6x, \qquad f_{xy}(x, y) = -6y, \qquad f_{yy}(x, y) = -6x.$$

Thus

$$H_f(x,y) = \begin{pmatrix} 6x & -6y \\ -6y & -6x \end{pmatrix}, \qquad H_f(0,0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Hence

$$\det(H_f(0,0)) = 0,$$

so the critical point $(0,0)$ is *degenerate*. Intuitively, this happens because the Taylor expansion of f at $(0,0)$ starts with cubic terms: the quadratic part vanishes completely.

Example 2: a non-degenerate critical point (a saddle)

Consider

$$g(u,v) = uv.$$

Step 1: critical point. The gradient is

$$\nabla g(u,v) = (g_u(u,v), g_v(u,v)) = (v,u).$$

Thus $\nabla g(0,0) = (0,0)$, so $(0,0)$ is a critical point.
Step 2: non-degeneracy. The second derivatives are

$$g_{uu} = 0, \qquad g_{uv} = 1, \qquad g_{vv} = 0,$$

so the Hessian matrix is constant:

$$H_g = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \det(H_g) = -1 \neq 0.$$

Therefore $(0,0)$ is a *non-degenerate* critical point (in fact, a saddle point). This is the local model that appears in the case $n = 2$ for the determinant function on $M(2)$ (restricted to diagonal matrices).

Geometric meaning of a critical point ("horizontal tangent plane")

Let $f : \mathbb{R}^k \to \mathbb{R}$ be smooth and let $x_0 \in \mathbb{R}^k$. The condition that x_0 is a critical point is

$$\nabla f(x_0) = 0 \qquad \Longleftrightarrow \qquad df_{x_0} = 0.$$

Geometrically, this means that the *first-order* (linear) change of f at x_0 vanishes in every direction.

(1) The tangent approximation. For a small vector $h \in \mathbb{R}^k$, Taylor's formula gives

$$f(x_0 + h) = f(x_0) + \nabla f(x_0) \cdot h + \frac{1}{2}\, h^T H_f(x_0)\, h + (\text{higher order terms}).$$

The term $\nabla f(x_0) \cdot h$ is the *first-order slope*. Thus if $\nabla f(x_0) = 0$, then

$$f(x_0 + h) = f(x_0) + \frac{1}{2}\, h^T H_f(x_0)\, h + \cdots,$$

so f does not change to first order when we move away from x_0; the first nontrivial change begins at second order (or at even higher order if the critical point is degenerate).

(2) The case $k = 2$: a surface in $\mathbb{R}^3$. If $f(x, y)$ is a function of two variables, its graph is the surface

$$S = \{(x, y, z) \in \mathbb{R}^3 : \; z = f(x, y)\}.$$

At a point (x_0, y_0) the tangent plane to S is

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Hence (x_0, y_0) is critical iff $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$, and then the tangent plane simplifies to

$$z = f(x_0, y_0),$$

which is a *horizontal plane*. This is the precise meaning of the phrase "the tangent plane is horizontal at a critical point."

(3) What you see on a contour plot. A contour plot draws level sets $f(x, y) = c$. If $\nabla f(x_0, y_0) \neq 0$, then nearby level curves are smooth and their normal direction is given by the gradient. If $\nabla f(x_0, y_0) = 0$, the point can become a special point of the level sets: for a non-degenerate minimum/maximum one typically sees small closed curves around (x_0, y_0), for a saddle one sees level curves crossing, and for degenerate critical points one can see multi-armed or "pinched" patterns (e.g. the monkey saddle).

Figures.

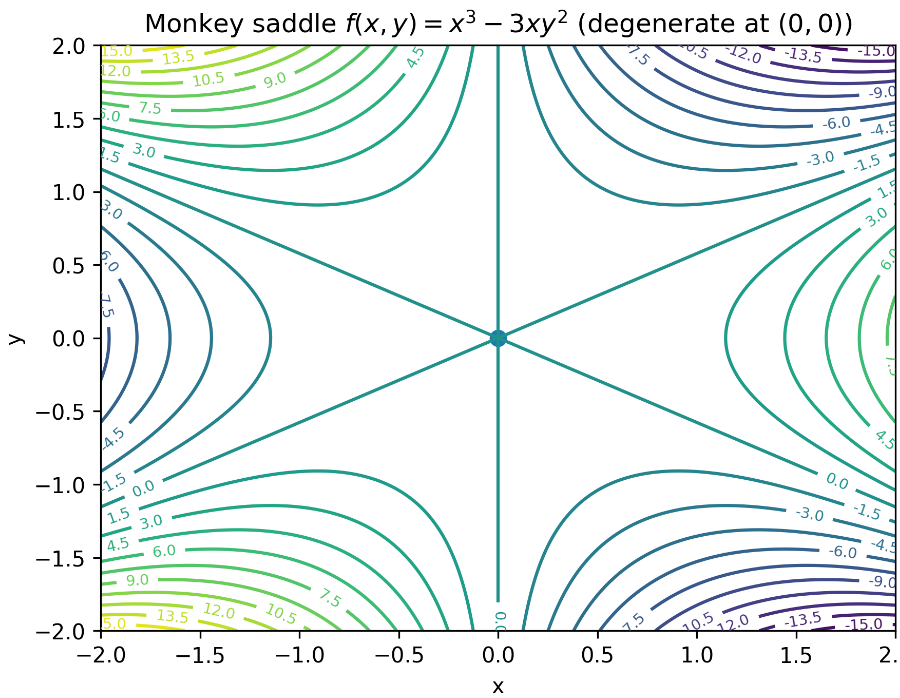


Figure 32: Contour plot of the monkey saddle $f(x, y) = x^3 - 3xy^2$: $(0, 0)$ is critical and degenerate (the quadratic part vanishes).

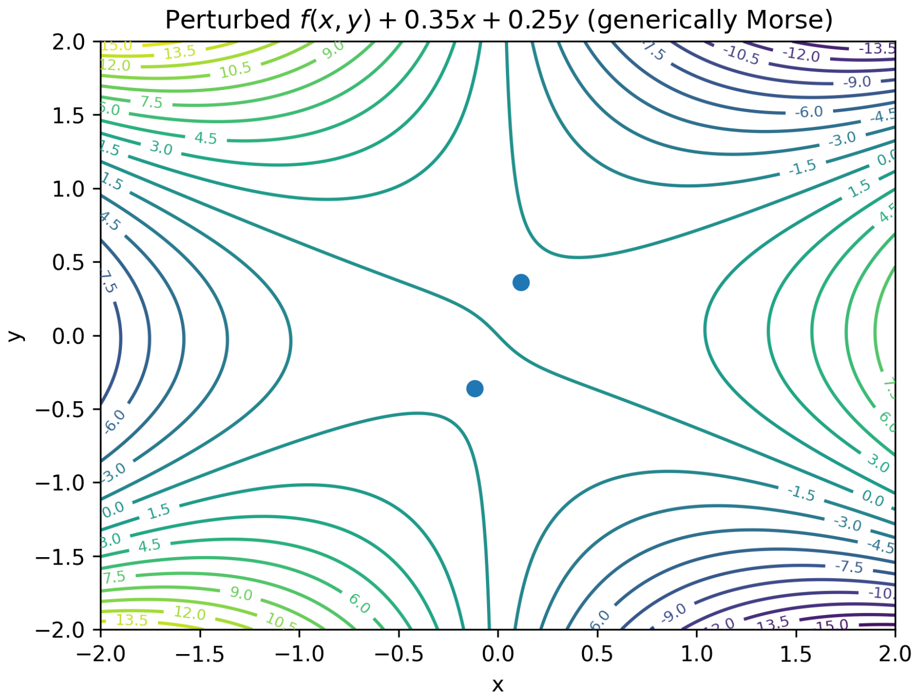


Figure 33: Generic linear perturbation $f_a = f + ax + by$ produces isolated non-degenerate critical points (a Morse function).

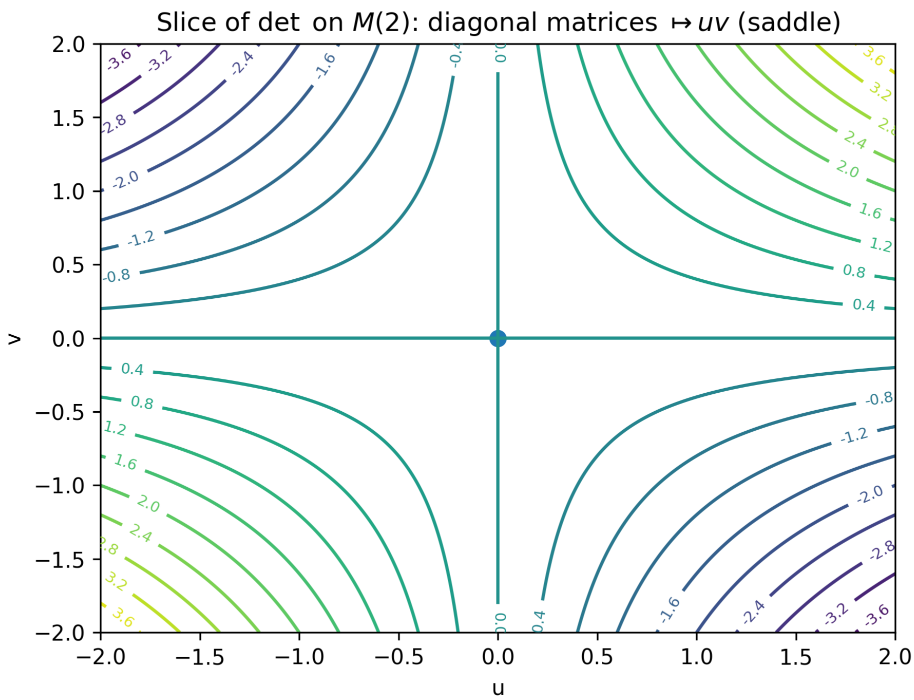


Figure 34: A diagonal slice of det on $M(2)$ equals uv and has a non-degenerate saddle critical point at $(0, 0)$.

Taylor expansion, saddles, and degeneracy (1D and 2D)

1D: critical points and the second derivative test. Let $f : \mathbb{R} \to \mathbb{R}$ be C^2 and let $x_0 \in \mathbb{R}$. A *critical point* is a point where

$$f'(x_0) = 0.$$

Taylor's theorem (with remainder) gives, for x near x_0,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2 + o\big((x - x_0)^2\big).$$

At a critical point $f'(x_0) = 0$, so

$$f(x) = f(x_0) + \frac{1}{2} f''(x_0)(x - x_0)^2 + o\big((x - x_0)^2\big).$$

Hence:

- If $f''(x_0) > 0$, then x_0 is a *strict local minimum.*
- If $f''(x_0) < 0$, then x_0 is a *strict local maximum.*
- If $f''(x_0) = 0$, the second derivative test is *inconclusive*; one must use higher-order terms or other methods.

1D: why x^2 and x^4 differ near 0. Consider $f(x) = x^2$ and $g(x) = x^4$ at 0:

$$f(0) = 0, \quad f'(0) = 0, \quad f''(0) = 2 \neq 0,$$

so the *first nonzero Taylor term* is quadratic:

$$f(x) = x^2.$$

For $g(x) = x^4$,

$$g(0) = 0, \quad g'(0) = 0, \quad g''(0) = 0, \quad g^{(3)}(0) = 0, \quad g^{(4)}(0) = 24 \neq 0,$$

so the first nonzero Taylor term appears at order 4:

$$g(x) = \frac{1}{4!} g^{(4)}(0) x^4 = x^4.$$

This explains the geometric behavior: x^4 is *much flatter near* 0 (since for small $|x|$ we have $|x|^4 \ll |x|^2$), even though it grows faster for large $|x|$.

2D: extended Taylor expansion and the Hessian. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be C^2 and let $p = (a, b)$. Write $h = (h_1, h_2) = (x - a, y - b)$. Then Taylor's theorem in $\mathbb{R}^2$ gives

$$f(p + h) = f(p) + \nabla f(p) \cdot h + \frac{1}{2} h^\top H_f(p)\, h + o\big(\|h\|^2\big),$$

where

$$\nabla f(p) = \begin{pmatrix} f_x(p) \\ f_y(p) \end{pmatrix}, \qquad H_f(p) = \begin{pmatrix} f_{xx}(p) & f_{xy}(p) \\ f_{yx}(p) & f_{yy}(p) \end{pmatrix},$$

and

$$h^\top H_f(p)\, h = f_{xx}(p)\, h_1^2 + 2f_{xy}(p)\, h_1 h_2 + f_{yy}(p)\, h_2^2.$$

Critical points and leading-order shape. A point p is a *critical point* iff

$$\nabla f(p) = 0.$$

At a critical point the linear term vanishes, and the first term controlling the local geometry is the quadratic form

$$Q(h) = \frac{1}{2}\, h^\top H_f(p)\, h.$$

This is the multivariable analogue of using $f''(x_0)$ in 1D.

Nondegenerate vs degenerate critical points.

- p is *nondegenerate* if the Hessian is invertible:

$$\det H_f(p) \neq 0.$$

- p is *degenerate* if the Hessian is singular:

$$\det H_f(p) = 0.$$

  In this case the quadratic approximation does *not* fully determine the local shape; higher-order terms are needed.

2D classification via the determinant test (nondegenerate case). At a critical point p, set

$$D = \det H_f(p) = f_{xx}(p) f_{yy}(p) - f_{xy}(p)^2.$$

Then:

- If $D > 0$ and $f_{xx}(p) > 0$, then p is a *strict local minimum.*
- If $D > 0$ and $f_{xx}(p) < 0$, then p is a *strict local maximum.*
- If $D < 0$, then $H_f(p)$ is *indefinite* and p is a *saddle point.*
- If $D = 0$, the test is *inconclusive* (the point is degenerate).

Saddles vs degeneracy: key examples.

(1) Nondegenerate saddle. Let

$$f(x, y) = x^2 - y^2.$$

Then $(0, 0)$ is critical and

$$H_f(0,0) = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}, \qquad \det H_f(0,0) = -4 < 0.$$

Hence $(0, 0)$ is a *nondegenerate saddle.* Geometrically, the surface bends upward in the x-direction and downward in the y-direction.

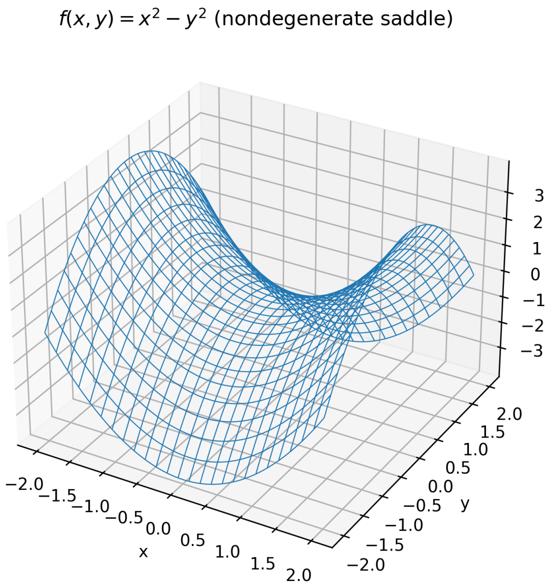
(2) Degenerate saddle. Let
$$g(x, y) = x^4 - y^4.$$
Then $(0, 0)$ is critical, and all second partial derivatives vanish at $(0, 0)$, so
$$H_g(0, 0) = 0, \qquad \det H_g(0, 0) = 0,$$
i.e. $(0, 0)$ is *degenerate.* Nevertheless it is still a *saddle* because
$$g(x, 0) = x^4 \geq 0, \qquad g(0, y) = -y^4 \leq 0,$$
so arbitrarily close to $(0, 0)$ one finds points where $g > 0$ and points where $g < 0$.

(3) Degenerate minimum (flat bottom). Let
$$h(x, y) = x^4 + y^4.$$
Then $(0, 0)$ is critical and degenerate ($H_h(0, 0) = 0$), but $h(x, y) \geq 0$ with equality only at $(0, 0)$, so $(0, 0)$ is a *(degenerate) strict local minimum.*

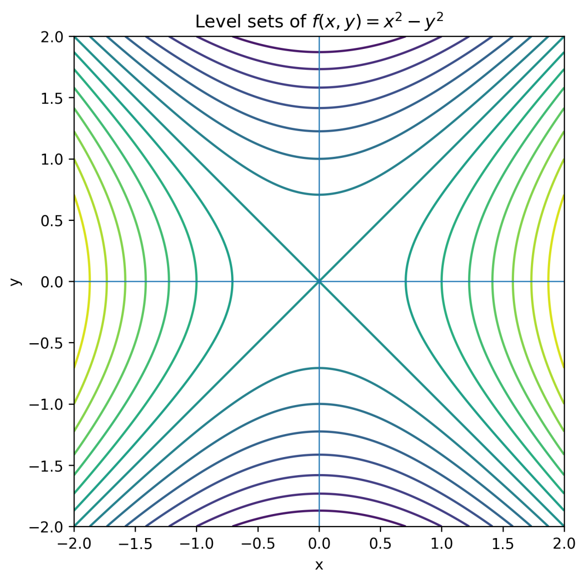
Moral.

- *Saddle* means: the function takes values both above and below $f(p)$ arbitrarily close to p.
- *Degenerate* means: the quadratic (Hessian) term is insufficient ($\det H_f(p) = 0$).
- Saddles can be nondegenerate (e.g. $x^2 - y^2$) or degenerate (e.g. $x^4 - y^4$).

Figures: nondegenerate vs degenerate saddle

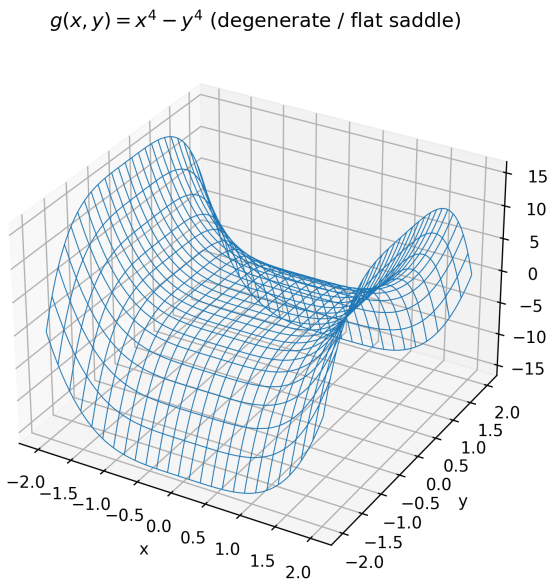


$f(x, y) = x^2 - y^2$: 3D surface (nondegenerate saddle).

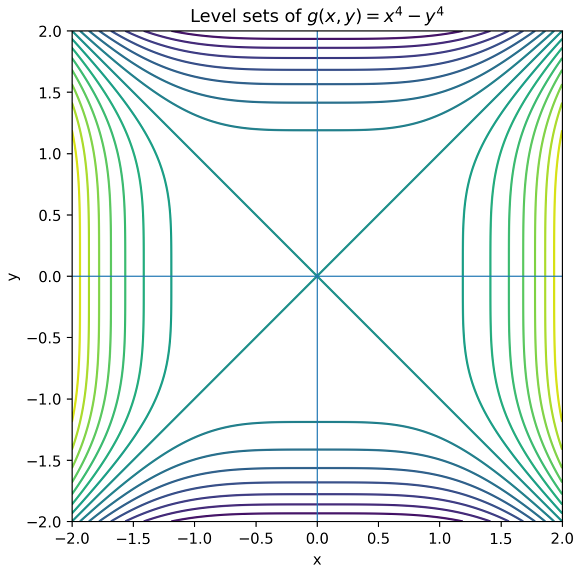


$f(x, y) = x^2 - y^2$: level sets (hyperbolas).

Figure 35: A nondegenerate saddle: curvature is already visible at quadratic order (Hessian indefinite).



$g(x, y) = x^4 - y^4$: 3D surface (degenerate / flat saddle).



$g(x, y) = x^4 - y^4$: level sets.

Figure 36: A degenerate saddle: the quadratic term vanishes (Hessian singular), so the saddle shape appears only in higher order.

Why $d(\det)_A(H) = \operatorname{tr}(\operatorname{adj}(A)^T H)$

Let $\det : M(n) \to \mathbb{R}$ be the determinant function. Fix $A \in M(n)$ and a direction $H \in M(n)$. Define a one-variable function

$$g(t) = \det(A + tH).$$

By definition of the directional derivative,

$$d(\det)_A(H) = g'(0).$$

Step 1: Multilinearity in columns. Write the columns of A as $a_1, \dots, a_n$ and the columns of H as $h_1, \dots, h_n$. Then

$$A + tH = [\, a_1 + th_1,\ a_2 + th_2,\ \dots,\ a_n + th_n \,].$$

Since det is multilinear in the columns,

$$\det(A + tH) = \det(a_1 + th_1, \dots, a_n + th_n)$$

$$= \det(a_1, \dots, a_n) + t \sum_{j=1}^{n} \det(a_1, \dots, h_j, \dots, a_n) + (\text{terms of order } t^2 \text{ and higher}).$$

Therefore the coefficient of t equals $g'(0)$, and hence

$$d(\det)_A(H) = g'(0) = \sum_{j=1}^{n} \det(a_1, \dots, h_j, \dots, a_n). \qquad (*)$$

Step 2: Laplace expansion along the replaced column. Fix an index j. Let $B = (a_1, \dots, h_j, \dots, a_n)$ be the matrix obtained by replacing the j-th column of A by h_j. Expand $\det(B)$ along the j-th column:

$$\det(B) = \sum_{i=1}^{n} B_{ij}\, C_{ij}(B),$$

where $C_{ij}(B) = (-1)^{i+j} \det(B_{\widehat{i},\widehat{j}})$ is the (i,j)-cofactor. Deleting row i and column j removes the replaced column, hence

$$B_{\widehat{i},\widehat{j}} = A_{\widehat{i},\widehat{j}} \qquad \Longrightarrow \qquad C_{ij}(B) = (-1)^{i+j} \det(A_{\widehat{i},\widehat{j}}) = C_{ij}(A).$$

Moreover $B_{ij} = (h_j)_i = H_{ij}$. Therefore

$$\det(a_1, \dots, h_j, \dots, a_n) = \det(B) = \sum_{i=1}^{n} H_{ij}\, C_{ij}(A).$$

Step 3: Sum over all j and identify the adjugate. Substituting the previous formula into $(*)$ gives

$$d(\det)_A(H) = \sum_{j=1}^{n} \sum_{i=1}^{n} H_{ij}\, C_{ij}(A) = \sum_{i,j} C_{ij}(A)\, H_{ij}.$$

Recall that the adjugate matrix satisfies

$$\operatorname{adj}(A)_{ji} = C_{ij}(A) \quad \text{(adjugate is the transpose of the cofactor matrix).}$$

Hence

$$\sum_{i,j} C_{ij}(A)\, H_{ij} = \sum_{i,j} \operatorname{adj}(A)_{ji}\, H_{ij}.$$

Finally, using the identity

$$\operatorname{tr}(X^T Y) = \sum_{i,j} X_{ij} Y_{ij},$$

we obtain

$$\operatorname{tr}\big(\operatorname{adj}(A)^T H\big) = \sum_{i,j} \big(\operatorname{adj}(A)^T\big)_{ij} H_{ij} = \sum_{i,j} \operatorname{adj}(A)_{ji} H_{ij}.$$

Therefore,

$$d(\det)_A(H) = \operatorname{tr}\big(\operatorname{adj}(A)^T H\big).$$

Why det is multilinear, and where the $t^2, t^3, \dots$ terms come from

Multilinearity (in columns). Write an $n \times n$ matrix by its columns

$$A = [c_1, \dots, c_n], \qquad c_j \in \mathbb{R}^n.$$

A map $F(c_1, \dots, c_n)$ is called *multilinear* if it is linear in each argument separately, i.e. for every fixed j and all vectors $u, v \in \mathbb{R}^n$ and scalars $\lambda \in \mathbb{R}$,

$$F(c_1, \dots, c_{j-1}, u{+}v, c_{j+1}, \dots, c_n) = F(c_1, \dots, c_{j-1}, u, c_{j+1}, \dots, c_n) + F(c_1, \dots, c_{j-1}, v, c_{j+1}, \dots, c_n),$$

and

$$F(c_1,\dots,c_{j-1},\lambda u,c_{j+1},\dots,c_n)=\lambda\,F(c_1,\dots,c_{j-1},u,c_{j+1},\dots,c_n).$$

For the determinant, this property follows from the Leibniz formula:

$$\det(A)=\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\prod_{i=1}^n a_{i,\sigma(i)}.$$

Fix a column index j. In each product $\prod_{i=1}^n a_{i,\sigma(i)}$, the permutation σ hits each column index exactly once, so *exactly one factor* in the product comes from the j-th column. Therefore each summand depends linearly on the entries of the j-th column, and so does the sum. Since j was arbitrary, det is linear in each column separately, i.e. det is multilinear.

Where the powers $t^2,t^3,\dots$ come from. Let $A,H\in M(n)$ and write their columns as

$$A=[a_1,\dots,a_n],\qquad H=[h_1,\dots,h_n].$$

Then

$$A+tH=[a_1+th_1,\dots,a_n+th_n].$$

Using multilinearity, we can expand $\det(A+tH)$ by distributing over each column (analogous to expanding a product). From each column a_j+th_j we choose either a_j or th_j. This yields a sum of 2^n determinants:

$$\det(A+tH)=\sum_{S\subset\{1,\dots,n\}}\det\big(b_1(S),\dots,b_n(S)\big),$$

where

$$b_j(S)=\begin{cases}th_j, & j\in S,\\ a_j, & j\notin S.\end{cases}$$

If $|S|=m$, then exactly m columns contribute a factor t, and by linearity we can factor these out:

$$\det\big(b_1(S),\dots,b_n(S)\big)=t^m\det\big(c_1(S),\dots,c_n(S)\big),$$

where $c_j(S)=h_j$ if $j\in S$ and $c_j(S)=a_j$ otherwise. Hence the expansion groups naturally by the size m of S:

$$\det(A+tH)=\det(A)+t\sum_{|S|=1}\det(\text{replace one column of }A\text{ by the corresponding column of }H)+t^2(\cdots)+$$

In particular, the terms with $t^2,t^3,\dots$ come from choosing th_j in two, three, $\dots$ columns respectively.

Why only the linear term matters for the derivative at $t=0$. If we write

$$\det(A+tH)=\det(A)+tL+t^2Q(t),$$

then

$$\frac{d}{dt}\Big|_{t=0}\det(A+tH)=L,$$

because $\frac{d}{dt}\big(t^2Q(t)\big)\big|_{t=0}=0$. Therefore $d(\det)_A(H)$ is exactly the coefficient of t in the multilinear expansion above.

Directional derivative of det for 3×3 and 4×4

3×3 case (fully expanded). Let

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}, \qquad H = \begin{pmatrix} p & q & r \\ s & t & u \\ v & w & x \end{pmatrix}.$$

Writing $A = [a_1, a_2, a_3]$ and $H = [h_1, h_2, h_3]$ by columns, multilinearity gives

$$d(\det)_A(H) = \frac{d}{dt}\Big|_0 \det(A + tH) = \det(h_1, a_2, a_3) + \det(a_1, h_2, a_3) + \det(a_1, a_2, h_3).$$

Expanding each term along its replaced column yields

$$d(\det)_A(H) = \sum_{i=1}^{3} C_{i1}(A)H_{i1} + \sum_{i=1}^{3} C_{i2}(A)H_{i2} + \sum_{i=1}^{3} C_{i3}(A)H_{i3} = \sum_{i,j=1}^{3} C_{ij}(A)H_{ij}.$$

The cofactors of A are

$$\begin{aligned} C_{11} &= ei - fh, & C_{12} &= fg - di, & C_{13} &= dh - eg, \\ C_{21} &= ch - bi, & C_{22} &= ai - cg, & C_{23} &= bg - ah, \\ C_{31} &= bf - ce, & C_{32} &= cd - af, & C_{33} &= ae - bd. \end{aligned}$$

Hence

$$\boxed{d(\det)_A(H) = (ei - fh)p + (fg - di)q + (dh - eg)r + (ch - bi)s + (ai - cg)t + (bg - ah)u + (bf - ce)}$$

4×4 case (column-replacement formula). Write $A = [a_1, a_2, a_3, a_4]$ and $H = [h_1, h_2, h_3, h_4]$. Then

$$\boxed{d(\det)_A(H) = \det(h_1, a_2, a_3, a_4) + \det(a_1, h_2, a_3, a_4) + \det(a_1, a_2, h_3, a_4) + \det(a_1, a_2, a_3, h_4).}$$

Equivalently,

$$d(\det)_A(H) = \sum_{i,j=1}^{4} C_{ij}(A)H_{ij}, \qquad C_{ij}(A) = (-1)^{i+j} \det(A_{\hat{i},\hat{j}}).$$

Example (diagonal A). If $A = \operatorname{diag}(d_1, d_2, d_3, d_4)$ and $H = (h_{ij})$, then

$$\boxed{d(\det)_A(H) = d_2 d_3 d_4\, h_{11} + d_1 d_3 d_4\, h_{22} + d_1 d_2 d_4\, h_{33} + d_1 d_2 d_3\, h_{44}.}$$

Visualizing det : $M(2) \cong \mathbb{R}^4 \to \mathbb{R}$ and its derivative via slices

A 2×2 matrix can be parametrized by $(a, b, c, d) \in \mathbb{R}^4$:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \det(A) = ad - bc.$$

Since the domain is 4-dimensional, we cannot plot det directly. Instead, we fix two coordinates and visualize 2-dimensional slices. For example, if b, c are fixed then det becomes a function of (a, d):

$$\det(A) = ad - (bc),$$

so its level sets in the (a, d)-plane are hyperbolas. Similarly, if a, d are fixed then

$$\det(A) = (ad) - bc,$$

and level sets in the (b, c)-plane are again hyperbolas. To visualize the *derivative* on a slice, we plot the gradient vector field of the sliced function (arrows give the direction of steepest increase *within the slice*).

Figures (determinant slices).

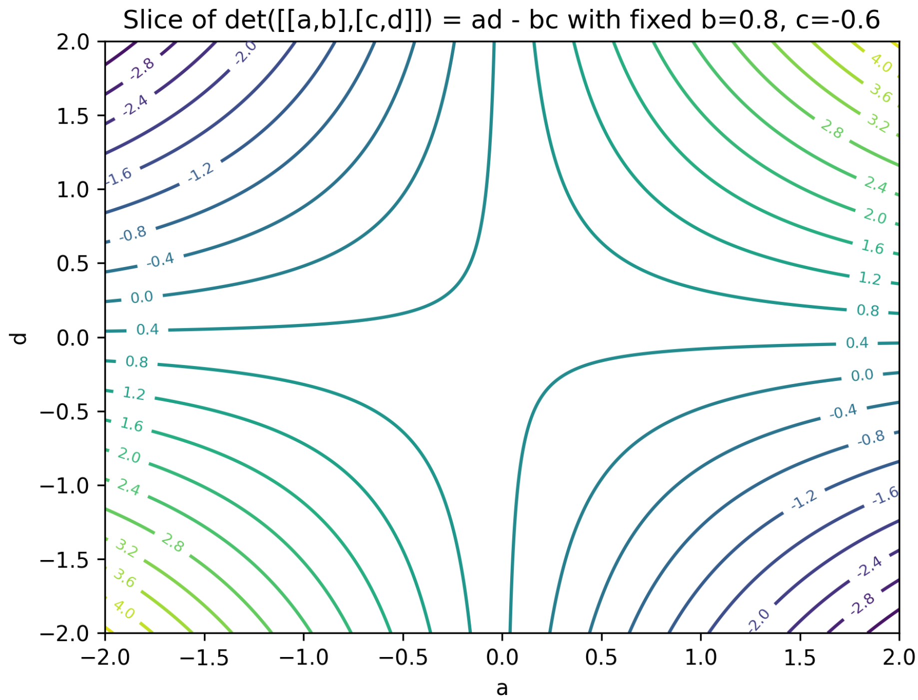


Figure 37: Contour plot of the slice with fixed b, c: $\det = ad - bc$ as a function of (a, d).

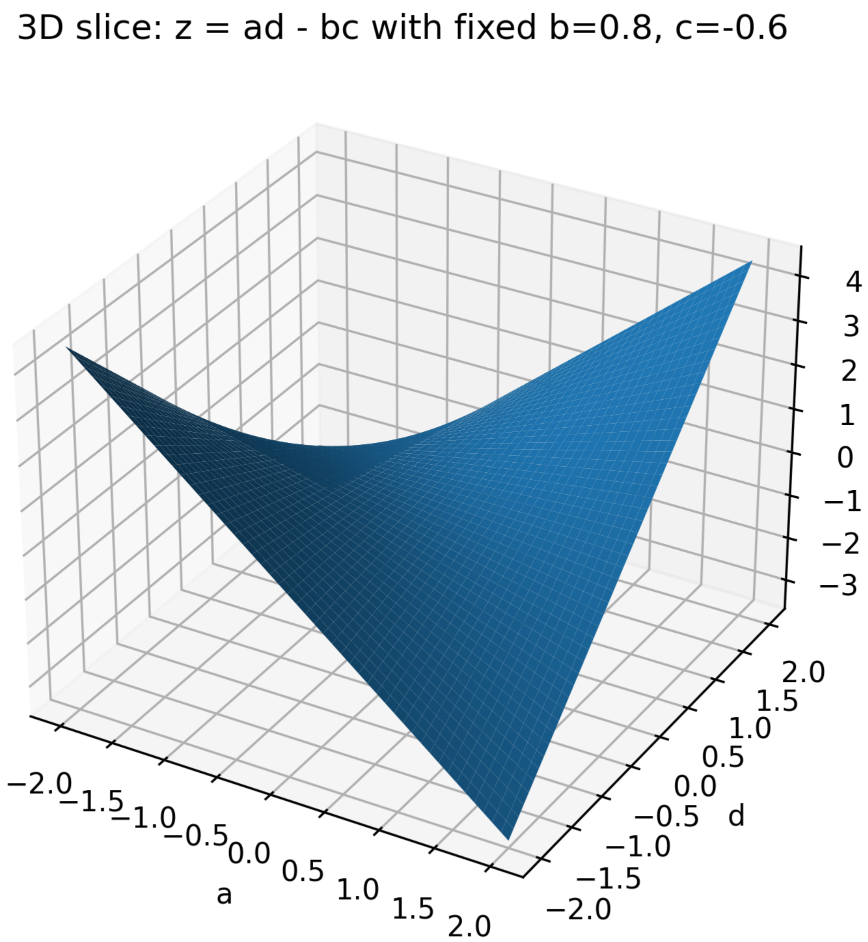


Figure 38: 3D surface slice (fixed b, c): $z = \det(a, d) = ad - bc$, a saddle shifted by $-bc$.

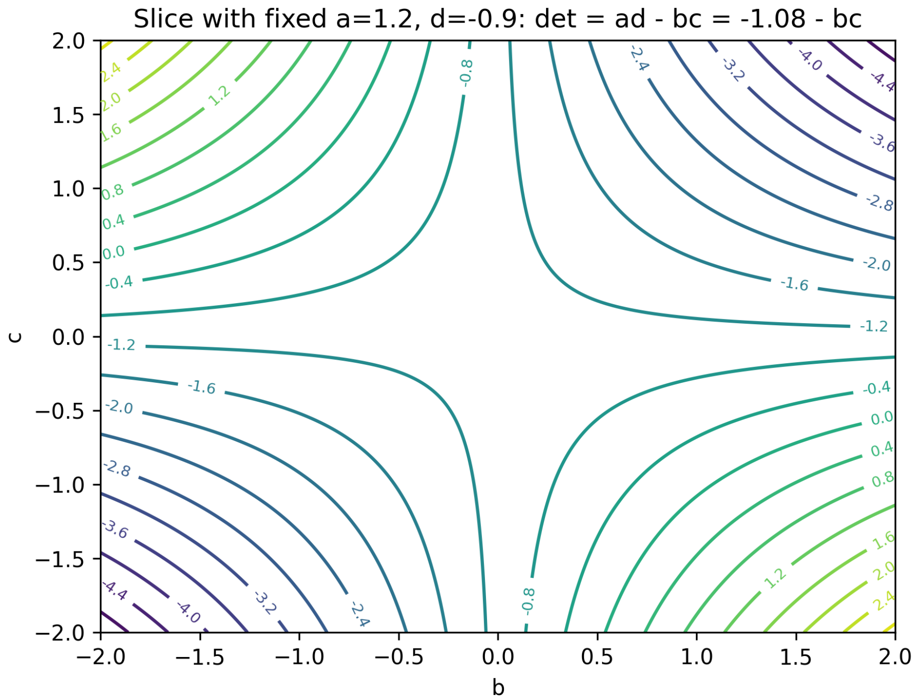


Figure 39: Contour plot of the slice with fixed a, d: $\det = ad - bc$ as a function of (b, c).

Figures (derivative on slices). On the (a, d)-slice (with fixed b, c), the sliced function is $F(a, d) = ad - (bc)$, hence

$$\nabla_{(a,d)} F(a, d) = (\partial_a F, \partial_d F) = (d, a).$$

On the (b, c)-slice (with fixed a, d), the sliced function is $G(b, c) = (ad) - bc$, hence

$$\nabla_{(b,c)} G(b, c) = (\partial_b G, \partial_c G) = (-c, -b).$$

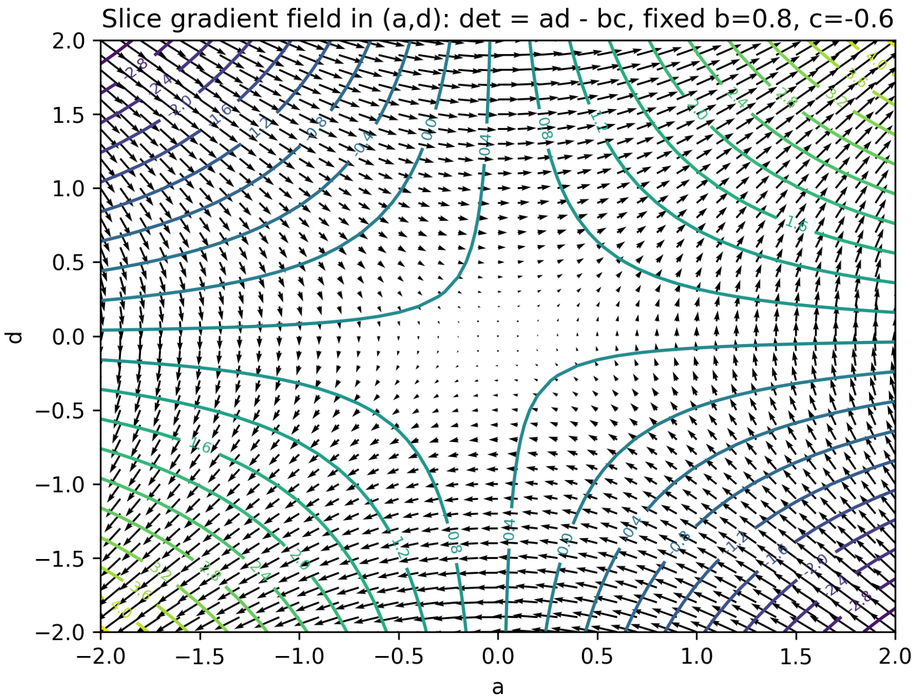


Figure 40: Gradient field on the (a, d)-slice (fixed b, c). Arrows show $\nabla_{(a,d)}(ad - bc) = (d, a)$.

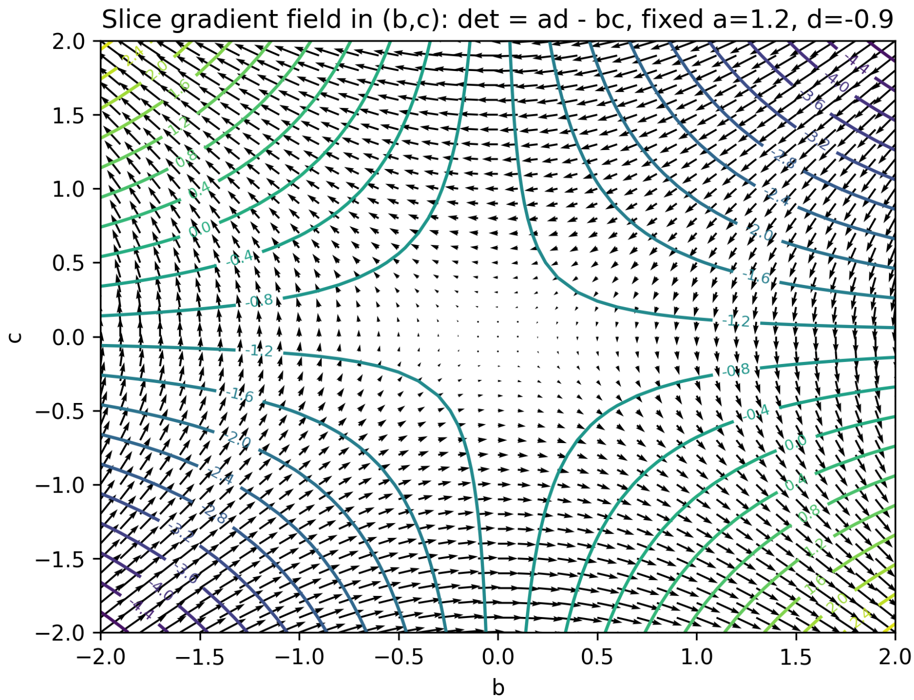


Figure 41: Gradient field on the (b, c)-slice (fixed a, d). Arrows show $\nabla_{(b,c)}(ad - bc) = (-c, -b)$.

Gradient vs. derivative on a slice of det

Identify $M(2) \cong \mathbb{R}^4$ via coordinates (a, b, c, d):

$$F(a,b,c,d) = \det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Since F has a 4-dimensional domain, we visualize it by fixing two variables and taking a 2-dimensional slice. Fix $b, c \in \mathbb{R}$. Then the restricted (slice) function is

$$f_{b,c}(a,d) := F(a,b,c,d) = ad - bc, \qquad (a,d) \in \mathbb{R}^2.$$

Derivative (differential) on the slice. The derivative of $f_{b,c}$ at a point (a,d) is the linear map

$$df_{b,c}\big|_{(a,d)} : \mathbb{R}^2 \to \mathbb{R},$$

defined by

$$df_{b,c}\big|_{(a,d)}(\Delta a, \Delta d) = \frac{\partial f_{b,c}}{\partial a}(a,d)\,\Delta a + \frac{\partial f_{b,c}}{\partial d}(a,d)\,\Delta d.$$

For $f_{b,c}(a,d) = ad - bc$ we have

$$\frac{\partial f_{b,c}}{\partial a}(a,d) = d, \qquad \frac{\partial f_{b,c}}{\partial d}(a,d) = a,$$

hence

$$\boxed{df_{b,c}\big|_{(a,d)}(\Delta a, \Delta d) = d\,\Delta a + a\,\Delta d.}$$

Gradient as a representation of the derivative. In $\mathbb{R}^2$ with the standard dot product, the differential can be written as

$$df_{b,c}\big|_{(a,d)}(\Delta a, \Delta d) = \nabla f_{b,c}(a,d) \cdot (\Delta a, \Delta d),$$

so the gradient vector is

$$\boxed{\nabla f_{b,c}(a,d) = \big(d, a\big).}$$

Therefore, the arrow field in the figure is a visualization of the derivative: each arrow at (a,d) points in the direction of fastest increase of $f_{b,c}$ within the (a,d)-slice.

Full 4-dimensional derivative (for completeness). For $F(a,b,c,d) = ad - bc$,

$$\nabla F(a,b,c,d) = \big(d, -c, -b, a\big),$$

and the differential at (a,b,c,d) is the linear map

$$\boxed{dF\big|_{(a,b,c,d)}(\Delta a, \Delta b, \Delta c, \Delta d) = d\,\Delta a - c\,\Delta b - b\,\Delta c + a\,\Delta d.}$$

The (a,d)-slice figure corresponds to restricting this differential to variations with $\Delta b = \Delta c = 0$.

Why $df_{(a,d)}(\Delta a, \Delta d) = \nabla f(a,d) \cdot (\Delta a, \Delta d)$?

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be differentiable and let $p = (a,d)$. The *differential* of f at p is the linear map (a linear functional)

$$df_p : \mathbb{R}^2 \to \mathbb{R}$$

characterized by the first–order Taylor approximation

$$f(p+h) = f(p) + df_p(h) + o(\|h\|) \qquad (h \to 0).$$

Writing $h = (\Delta a, \Delta d)$ and using partial derivatives, one obtains the explicit formula

$$df_{(a,d)}(\Delta a, \Delta d) = \frac{\partial f}{\partial x}(a,d)\,\Delta a + \frac{\partial f}{\partial y}(a,d)\,\Delta d.$$

Now introduce the *gradient* (with respect to the standard dot product on $\mathbb{R}^2$)

$$\nabla f(a,d) = \left(\frac{\partial f}{\partial x}(a,d),\ \frac{\partial f}{\partial y}(a,d)\right).$$

Then the same expression can be rewritten as a dot product:

$$\frac{\partial f}{\partial x}(a,d)\,\Delta a + \frac{\partial f}{\partial y}(a,d)\,\Delta d = \left(\frac{\partial f}{\partial x}(a,d), \frac{\partial f}{\partial y}(a,d)\right) \cdot (\Delta a, \Delta d) = \nabla f(a,d) \cdot (\Delta a, \Delta d).$$

Hence

$$df_{(a,d)}(\Delta a, \Delta d) = \nabla f(a,d) \cdot (\Delta a, \Delta d).$$

Conceptual reason. The differential df_p is a covector (linear functional). In Euclidean space with the standard inner product $\langle \cdot, \cdot \rangle$, every covector L can be represented uniquely as

$$L(h) = \langle v, h \rangle \quad \text{for a unique vector } v.$$

The gradient is precisely this representing vector: it is defined by

$$df_p(h) = \langle \nabla f(p), h \rangle \qquad \text{for all } h \in \mathbb{R}^2.$$

For the standard inner product, $\langle u, v \rangle = u \cdot v$, giving the stated formula.